Network Layer

October 5, 2026

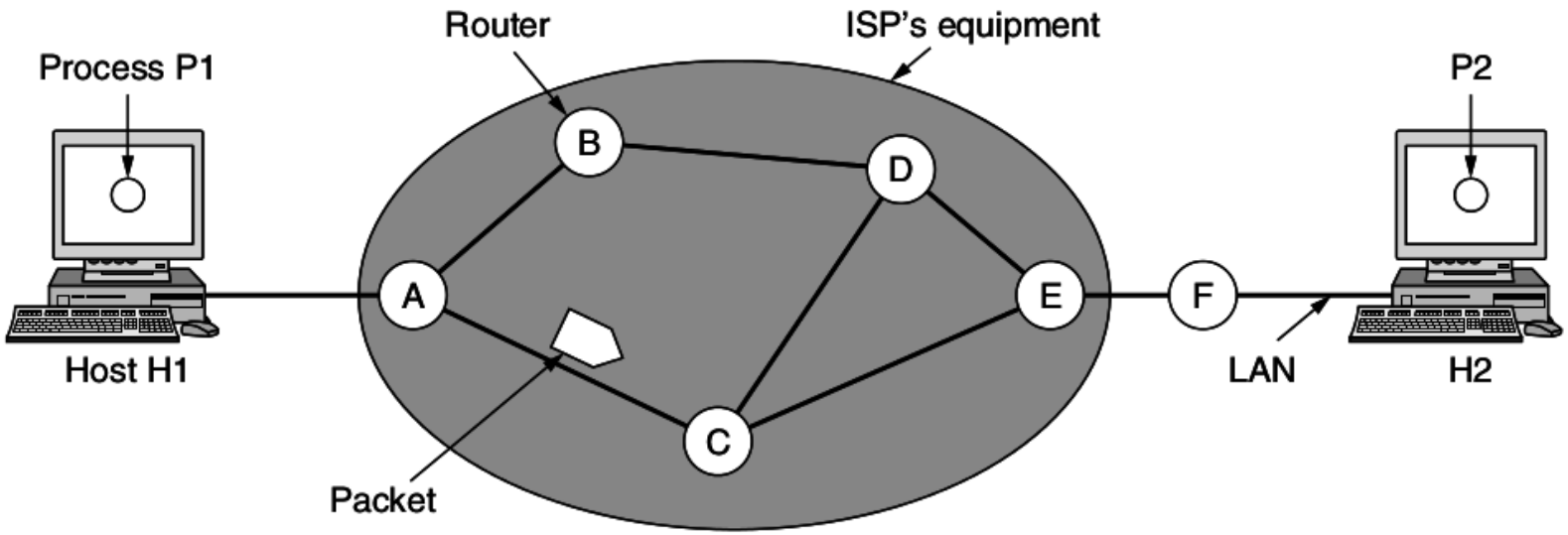
Goals

- Getting packets from the source all the way to the destination.
- Getting to the destination may require making many hops at intermediate routers along the way.

How to achieve this?

- To achieve its goals, N/W layer must know about the topology of the network (i.e., the set of all routers and links) and choose appropriate paths through it.
- Routes chosen should avoid overloading some of the communication lines and routers while leaving others idle.

Store and Forward Packet Switching



- A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the ISP.
- The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum.
- Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.

Services provided to Transport Layer

- The network layer is the lowest layer that carries data beyond a single network: from the source host to the destination host, across whatever networks lie between.
- The network layer provides services to the transport layer at the network layer/transport layer interface.
- Question/Dilemma: Should N/W layer provide connection oriented service or connectionless service.

Connection less service

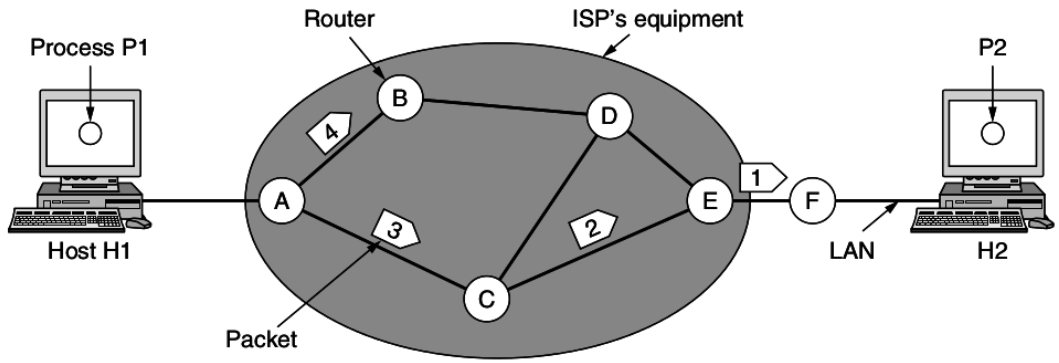
- Argument: Routers' job is moving packets around and nothing else.
- The network is inherently unreliable, no matter how it is designed. Therefore, the hosts should accept this fact and do error control (i.e., error detection and correction) and flow control themselves.
- Should comprise SEND PACKET and RECEIVE PACKET primitives, and each packet must carry the full destination address, because each packet sent is carried independently of its predecessors, if any.

Connection oriented service

- Argument: Success of telephone network. Reliable and provides QoS.
- X.25, Frame Relay (CO) Vs. Internet (CL).

Implementation of Connection Less Service

- Packets are injected into the network individually and routed independently of each other. No advance setup is needed.
- Packets are frequently called datagrams (in analogy with telegrams) and the network is called a datagram network.
- The algorithm that manages the tables and makes the routing decisions is called the routing algorithm.



A's table (initially)

Dest.	Line
A	–
B	B
C	C
D	B
E	C
F	C

A's table (later)

A	–
B	B
C	C
D	B
E	B
F	B

C's table

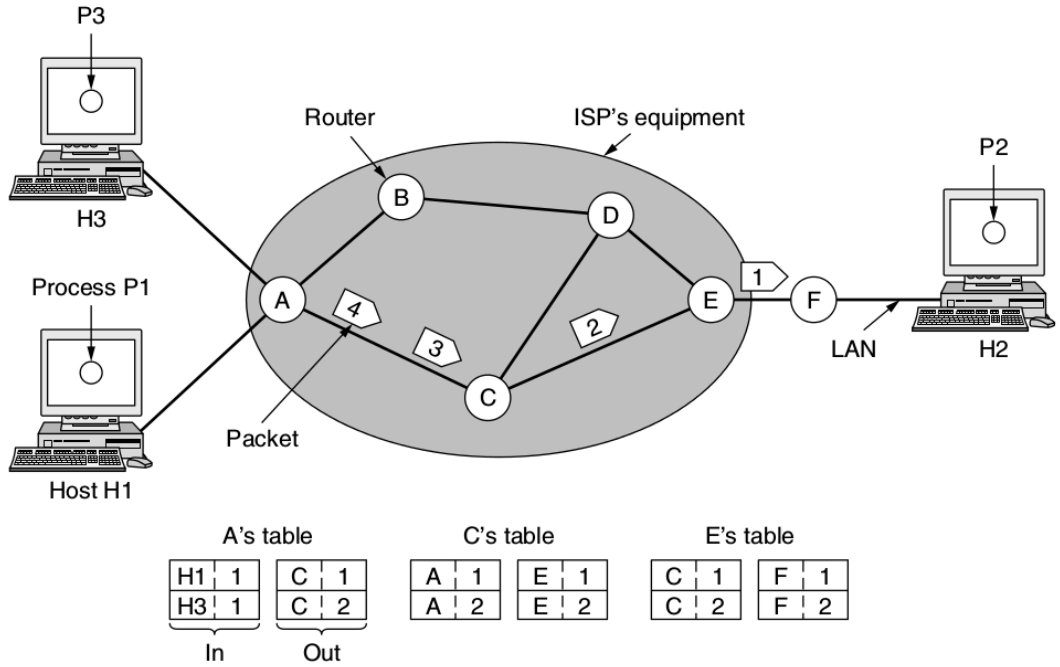
A	A
B	A
C	–
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

Implementation of Connection Oriented Service

- Path (VC) from the source router to destination router must be established before any data packets can be sent. Each packet carries an identifier telling which virtual circuit it belongs to.
- *A* assigns a different connection identifier to the outgoing traffic for the second connection. Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets.



VC & Datagram Networks

- Resources (e.g., buffers, bandwidth, and CPU cycles) can be reserved in advance, when the connection is established.

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Which model did the Internet choose?

- The Internet is a **datagram (connectionless)** network: IP gives a best-effort service and pushes reliability and flow control out to the end hosts (TCP).
- Why: no per-connection state in the core, so routers stay simple and the network rides out router failures – the "effect of router failures" row above – with packets rerouting independently around damage.
- Consequence: every router must decide, *per packet*, which outgoing line to use. **How it decides is the routing algorithm**.

Routing Algorithms

- The **routing algorithm** is the part of the network-layer software that determines the paths (routes) packets take through the network.
- A router can be thought of as running two processes:
  1. **Forwarding**: handle each packet as it arrives by looking up its outgoing line in the routing table. Local, per-packet, fast — the *data plane*.
  2. **Routing**: compute the best paths to all destinations and fill in / update the routing table accordingly. Network-wide, runs in the background — the *control plane*. **This is the routing algorithm.**
- **In short: forwarding consults the table; routing builds it.**
- Desirable properties: correctness, simplicity, robustness, stability, fairness, and efficiency.

Classes of Routing Algorithms

- **Non-adaptive (static) routing:** routes are computed in advance, offline, and downloaded to the routers when the network is booted. They do not react to changes in topology or traffic.
- **Adaptive (dynamic) routing:** routes change to reflect changes in the topology, and sometimes changes in the traffic as well.

Adaptive (dynamic) algorithms differ in three respects:

1. where they get their information (from adjacent routers, or from all routers),
2. when they change the routes (periodically, or on topology/load changes), and
3. what metric is used for optimization (hop count, physical distance, delay, . . .).

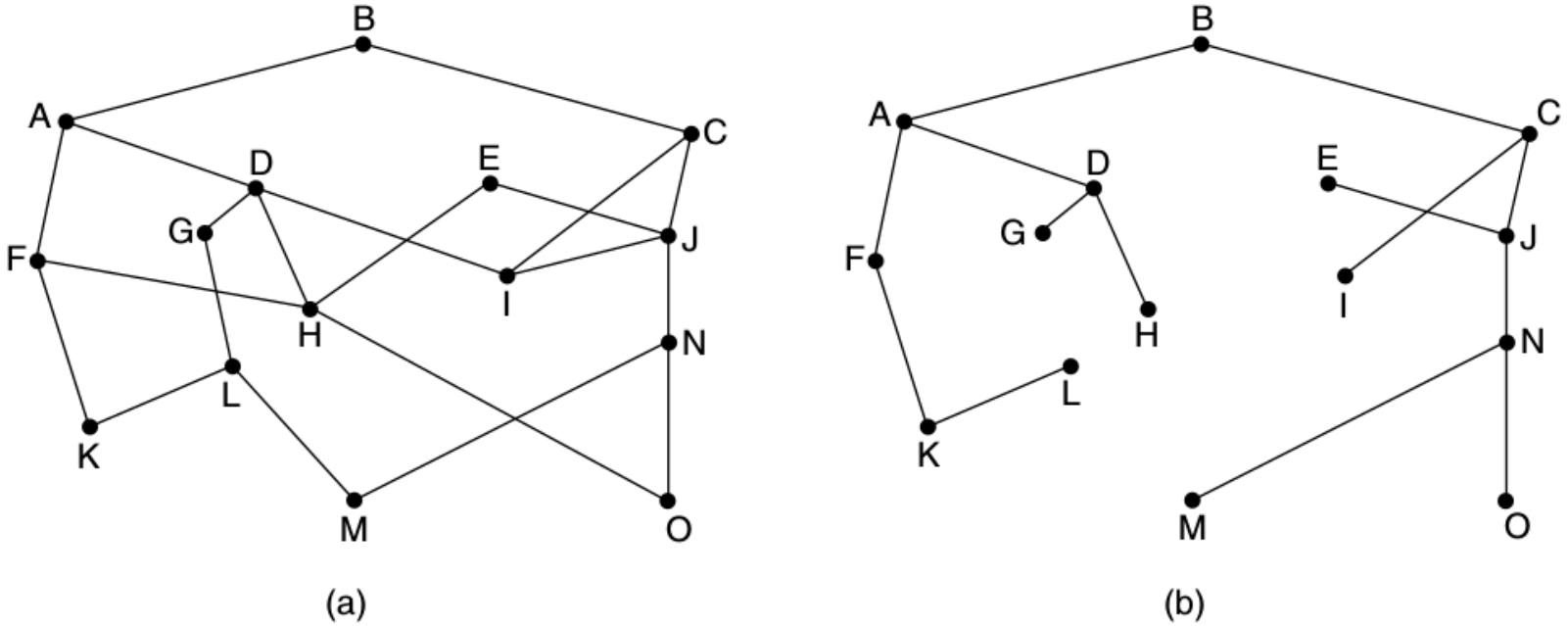
Optimality Principle

- It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.

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- To see this, call the part of the route from I to J r_1 and the rest of the route r_2.
- If a route better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K, contradicting our statement that r_1 r_2 is optimal.

Optimality Principle



- Based on optimality principle, the set of optimal routes from all sources to a given destination form a tree rooted at the destination **(sink tree)** where the distance metric is the number of hops.
- The **goal of all routing algorithms** is to discover and use the sink trees for all routers.

Optimality Principle

- A sink tree is not necessarily unique; other trees with the same path lengths may exist.
- Since a sink tree is indeed a tree, it does not contain any loops, so each packet will be delivered within a finite and bounded number of hops.

Shortest Path Algorithm

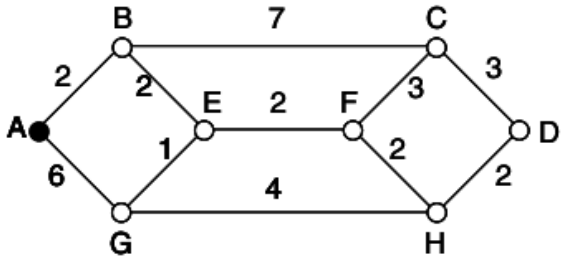
Lets us compute optimal paths given a complete picture of the network.

- Dijkstra's algorithm finds the shortest paths from one source to all destinations.
- Each node carries a label (d, p): d is its distance from the source along the best path known so far, and p is the previous node on that path.
- A label is *tentative* (a current guess) or *permanent* (known shortest).
- Initially the source is $(0, -)$ and permanent; every other node is $(\infty, -)$ and tentative.

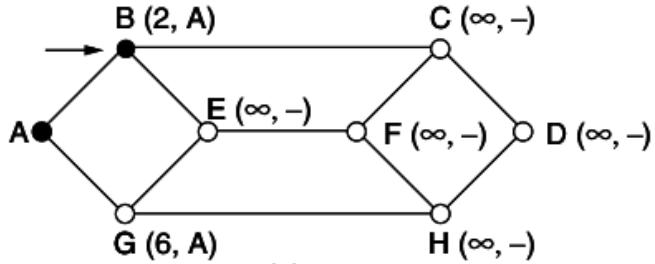
Shortest Path Algorithm

- **Select:** among all tentative nodes, make the one with the smallest d permanent i.e. a node with non-negative link costs, no later path can beat it. Call it the working node.
- **Relax:** for each tentative neighbour of the working node, if (working node's d + link cost) $<$ the neighbour's current d, set the neighbour's label to this distance with the working node as predecessor.
- Repeat Select/Relax until all nodes are permanent. The predecessor entries then trace each shortest path back to the source.

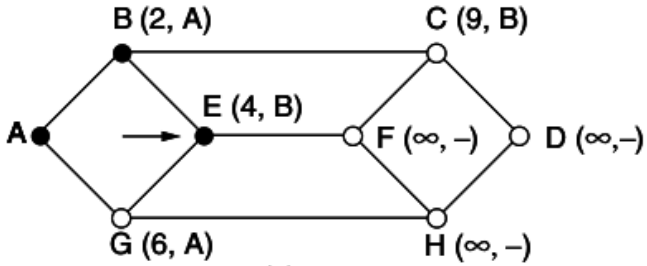
Shortest Path Algorithm



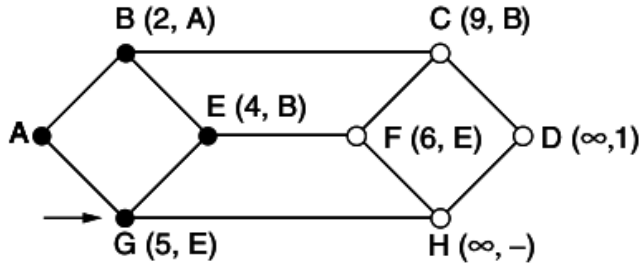
(a)



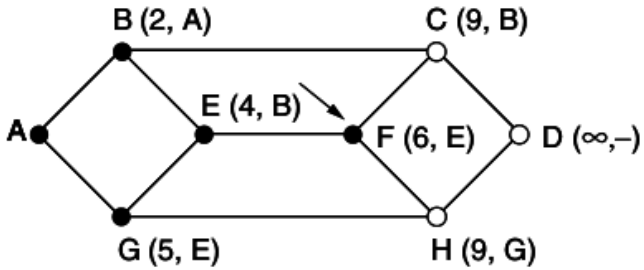
(b)



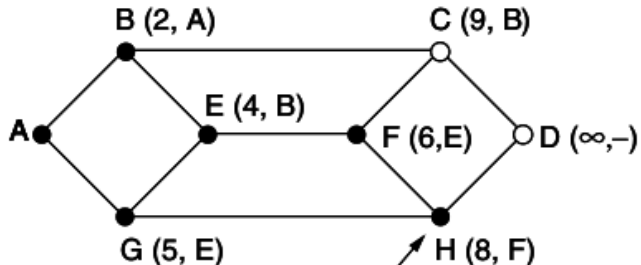
(c)



(d)



(e)



(f)

Flooding

Flooding is a **static** algorithm that needs no knowledge of the network at all: no topology, no routing tables, no setup.

- Every incoming packet is sent out on every outgoing line except the one it arrived on.
- It is remarkably robust. If any path to the destination exists, flooding finds it, and because it tries all paths in parallel, one copy travels the shortest path. It is therefore used for broadcasting and as a baseline against which other algorithms are measured.
- The price is duplication: an unbounded number of duplicate packets. Two ways to contain it:
  - a **hop counter** in each packet, decremented at each hop and discarded at zero; ideally initialised to the source-to-destination path length, or, if unknown, to the network diameter (the worst case);
  - have routers **track which packets they have already flooded** (by source and sequence number) and never forward one twice.

Distance Vector Routing

- Each router keeps a table listing, for every destination, its best known distance and the outgoing link to use. The row of distances alone is the **distance vector**.
- Routers **periodically** send their distance vector to their neighbours. On receiving the neighbours' vectors, a router re-estimates its distance to each destination D:

$$\mathrm{dist}(D) \;=\; \min_{\text{neighbour } N} \big(c(N) + \mathrm{dist}_N(D) \big),$$

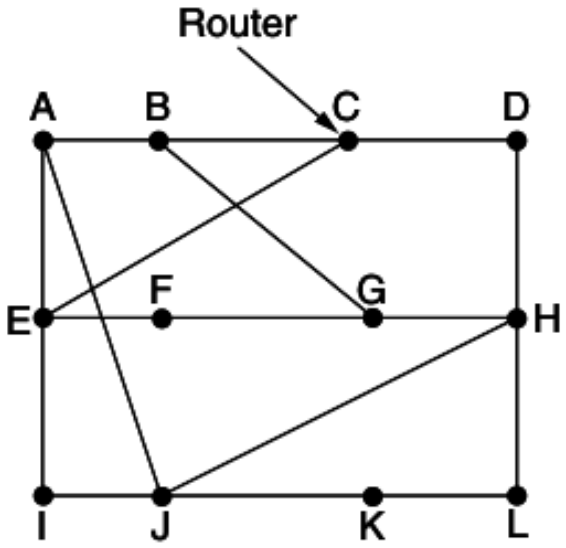
where $c(N)$ is the link cost to neighbour N (here, the measured delay) and $\mathrm{dist}_N(D)$ is N's advertised distance to D. The neighbour attaining the minimum becomes the outgoing link.

Distance Vector Routing

- This is the distributed form of the **Bellman-Ford** algorithm: routers recompute continuously as vectors arrive. Eventually every router learns the best link to each destination (convergence).
- It was the original ARPANET routing algorithm and was later used in the Internet as **RIP** (Routing Information Protocol).

DV Routing

Suppose that J has measured or estimated its delay to its neighbors, A, I, H, and K, as 8, 10, 12, and 6 msec, respectively.



New estimated delay from J

To	A	I	H	K	Line
A	0	24	20	21	8 A
B	12	36	31	28	20 A
C	25	18	19	36	28 I
D	40	27	8	24	20 H
E	14	7	30	22	17 I
F	23	20	19	40	30 I
G	18	31	6	31	18 H
H	17	20	0	19	12 H
I	21	0	14	22	10 I
J	9	11	7	10	0 –
K	24	22	22	0	6 K
L	29	33	9	9	15 K

JA delay is 8 | JI delay is 10 | JH delay is 12 | JK delay is 6

Vectors received from J's four neighbors

New routing table for J

Count to Infinity Problem

- The settling of routes to best paths across the network is called convergence.
- DV reacts rapidly to good news, but leisurely to bad news.

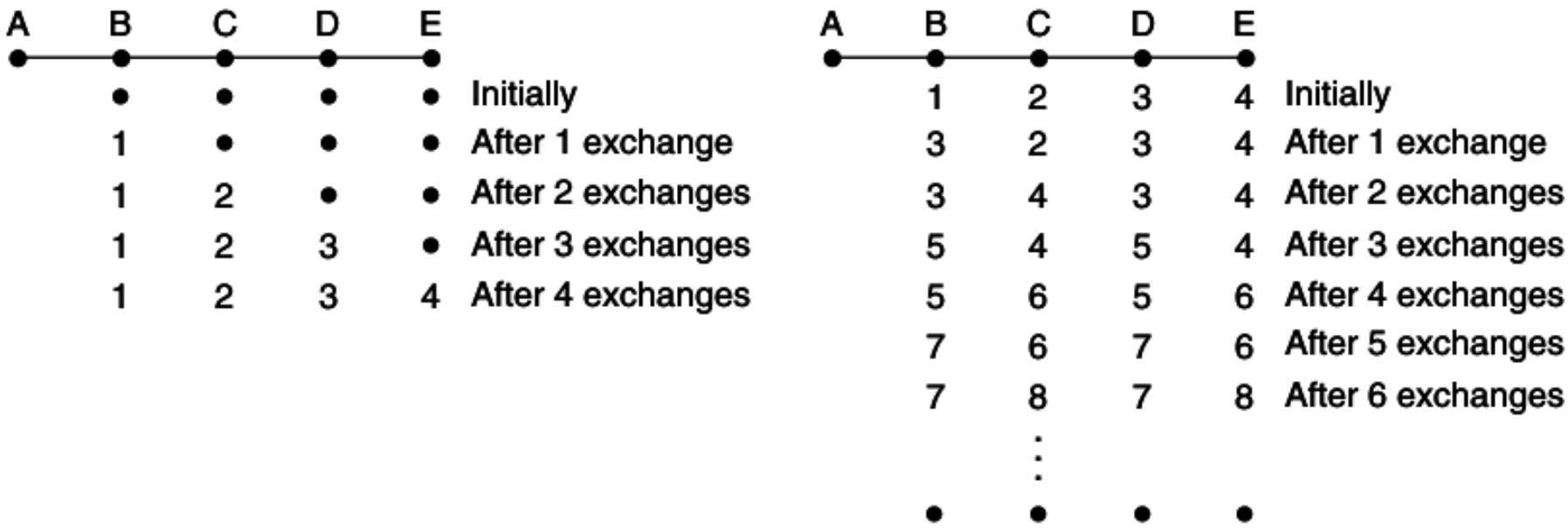


Figure: **A** was down then up, **A** was up then down

Link State Routing

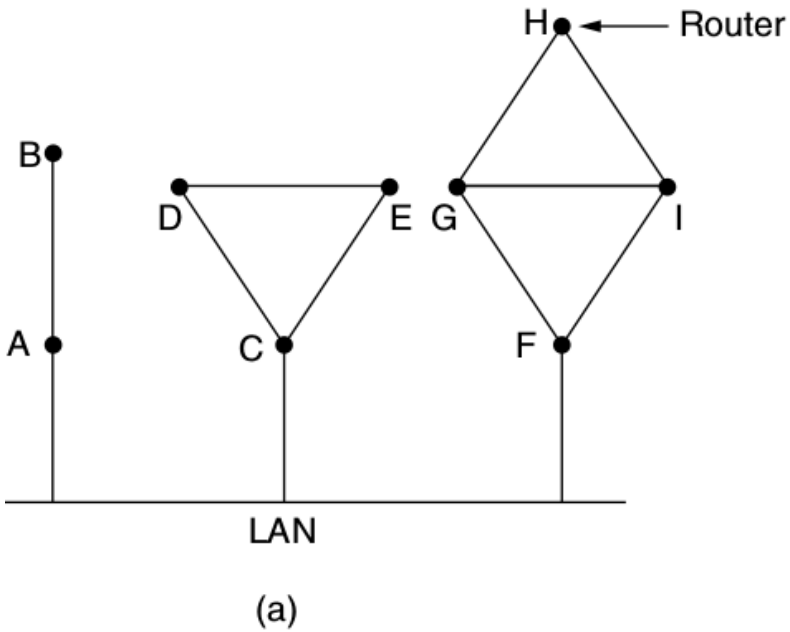
Distance Vector Routing takes too long to converge.
In link state routing each router must do the following:

- **Discover its neighbors** and learn their network addresses.
- **Set the distance or cost metric** to each of its neighbors.
- **Construct a packet** telling all it has just learned.
- Send this packet to and receive packets from **all other routers**.
- **Compute the shortest path** to every other router.

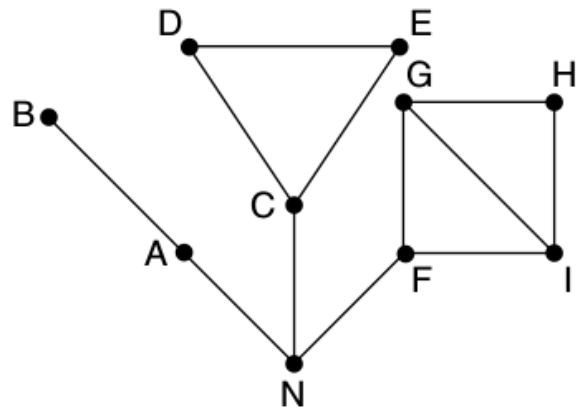
In effect, the complete topology is distributed to every router and Dijkstra's algorithm can be run at each router to find the shortest path to every other router.

Link State Routing: Discovering Neighbors

- Sends a special **HELLO** packet on each point-to-point line. The router on the other end is expected to send back a reply giving its name.
- The router names must be globally unique.

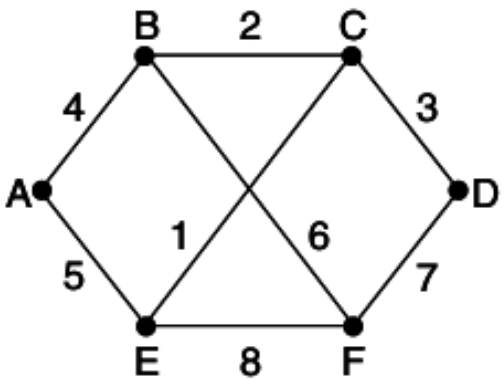


(a)



(b)

Link State Routing: Building Link State Packets



(a)

Link State Packets

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

(b)

Link State Routing: Distributing Link State Packets

The fundamental idea is to use flooding to distribute the link state packets to all routers. To keep the flood in check, each packet contains a sequence number that is incremented for each new packet sent.

Routers keep track of all the (source router, sequence) pairs they see. The Age field is decremented by each router during the initial flooding process, to make sure no packet can get lost and live for an indefinite period of time.

			Send flags			ACK flags			
Source	Seq.	Age	A	C	F	A	C	F	Data
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Figure: The packet buffer for router B

Link State Routing: Computing Routes

Once a router has accumulated a full set of link state packets, it can construct the entire network graph because every link is represented.

Dijkstra's algorithm can be run locally to construct the shortest paths to all possible destinations.

Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS) are some examples.

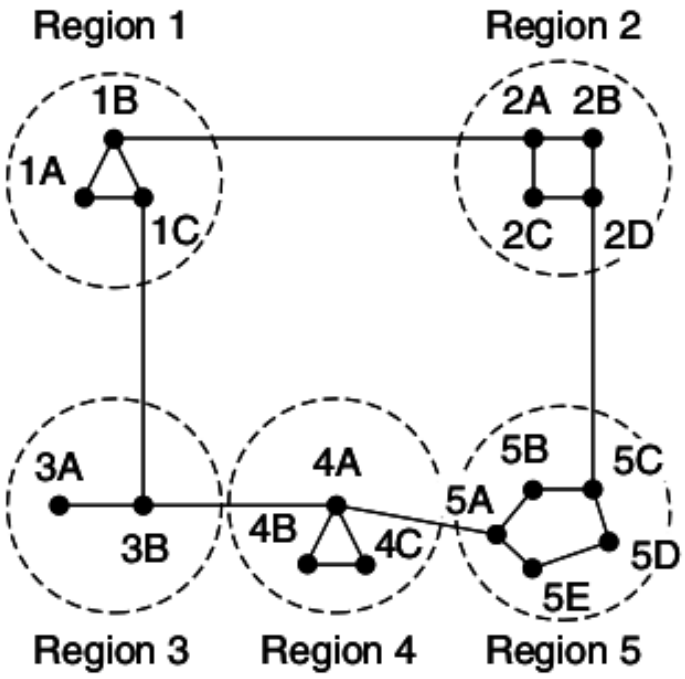
Hierarchical Routing

Can you guess why there is a need for hierarchical routing?

Hierarchical Routing

- As networks grow in size, the router routing tables grow proportionally.
- More router memory, more CPU time and more bandwidth is needed to send status reports about them.
- The network may grow to the point where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically, similar to a telephone network.

Hierarchical Routing



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Broadcast Routing

- Sending a packet to all destinations at once is called **broadcasting**.
- Several methods, from the simplest to the most refined:
  1. **Separate packet per destination:** wasteful, and the source must know every address.
  2. **Flooding:** reaches everyone with no address list, but generates many duplicates.
  3. **Multidestination routing:** one packet carries a list (or bit map) of destinations. At each router the destinations are grouped by outgoing line, and a separate copy is sent on each line carrying only its own group. The list splits at every hop, so each copy ends up carrying a single destination.
  4. **Spanning-tree (sink-tree) broadcast:** forward a broadcast on every tree link except the one it arrived on, so each link carries it once. Most efficient, but the tree must first be built and each router told which of its links are on it.
  5. **Reverse path forwarding (RPF):** approximates the spanning tree without any router knowing it (next).

Reverse Path Forwarding (RPF)

A broadcast originates at source S; a copy arrives at router R on link L.

- R checks its ordinary routing table: is L the link it would normally use to send *towards* S?
- **If yes,** the copy most likely travelled the shortest path from S to R, so it is the first (and best) copy to arrive. R accepts it and forwards it on all other links.
- **If no,** the copy reached R by a longer path and is discarded as a likely duplicate.
- The test reuses the route R already holds for S, so no broadcast tree has to be built or stored. RPF thus approximates spanning-tree broadcast using only ordinary routing information.
- "Most likely" rather than "certainly": the route *to* S is a proxy for the reverse route *from* S, and the two coincide only when link costs are symmetric.

Reverse Path Forwarding

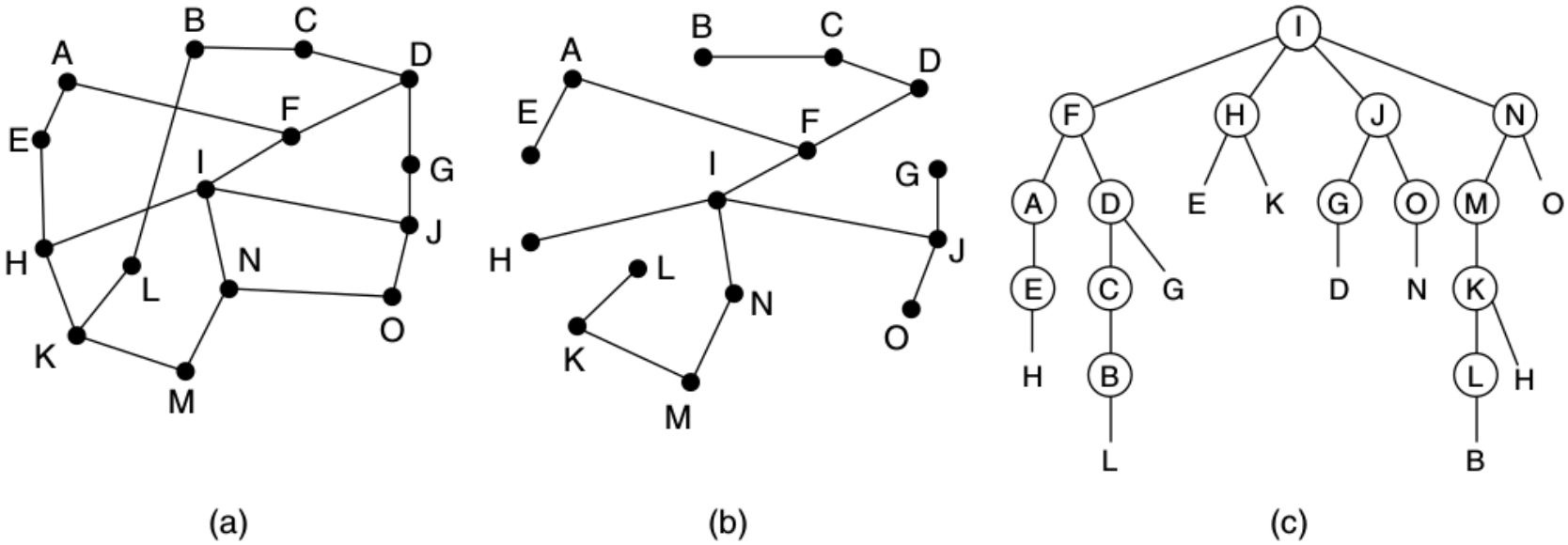


Figure 5-15. Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

After 5 hops and 24 packets, the broadcasting terminates.
4 hops and 14 packets would have been sufficient had the sink tree been followed exactly

Reverse path forwarding:

- The principal advantage of reverse path forwarding is that it is efficient while **being easy to implement**.
- It sends the broadcast packet over each link only once in each direction, **just as in flooding**, yet it requires only that routers know how to reach all destinations, without needing to remember sequence numbers (or use other mechanisms to stop the flood) or list all destinations in the packet.

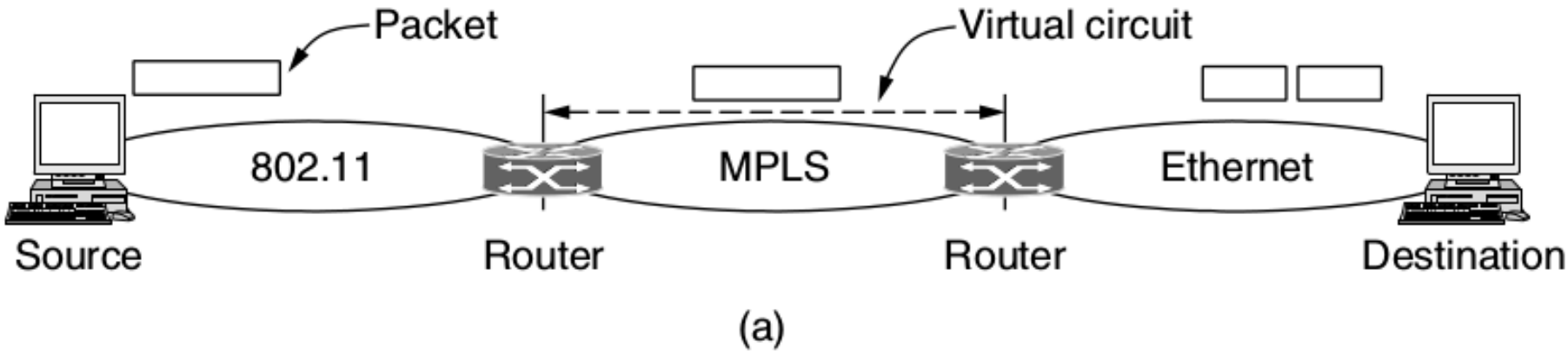
Internetworking

- Routing so far assumed one uniform network. The Internet is really many different networks joined together, and they are not all alike.
- What issues arise when two or more such networks are connected to form an **internetwork**, or more simply an **internet**?

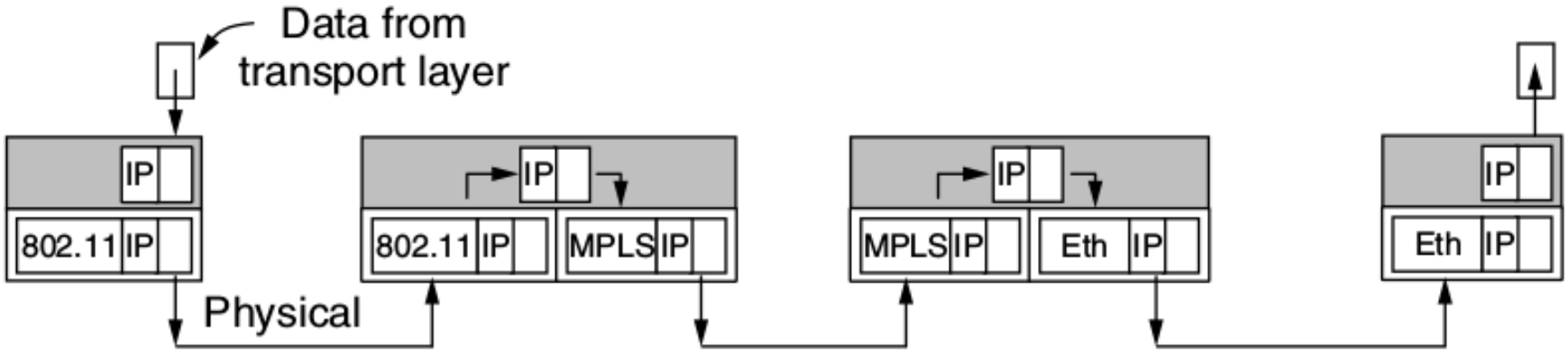
How networks differ?

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

How networks can be connected?



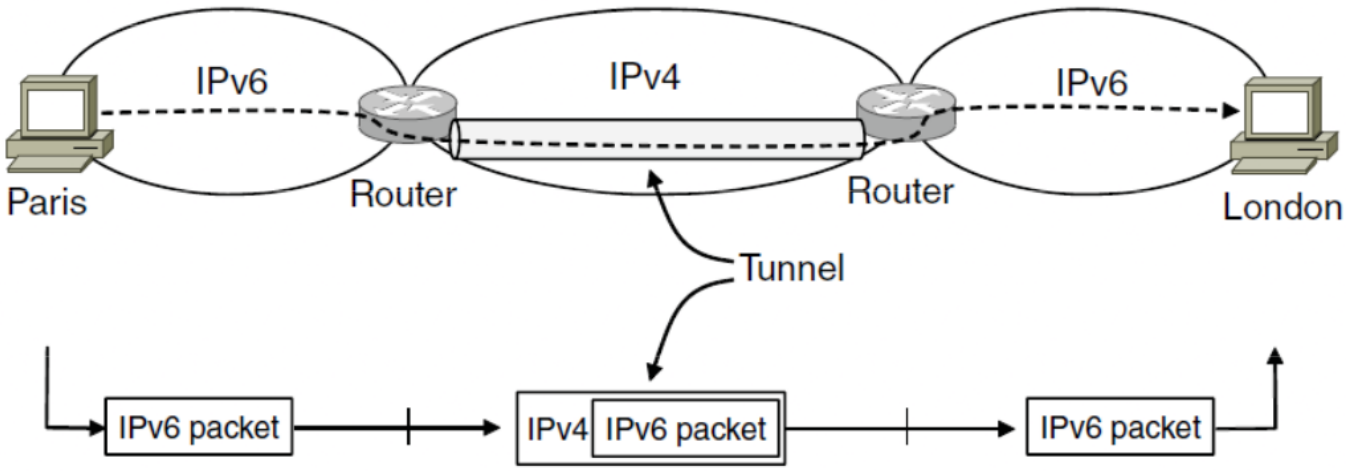
(a)



(b)

Tunneling

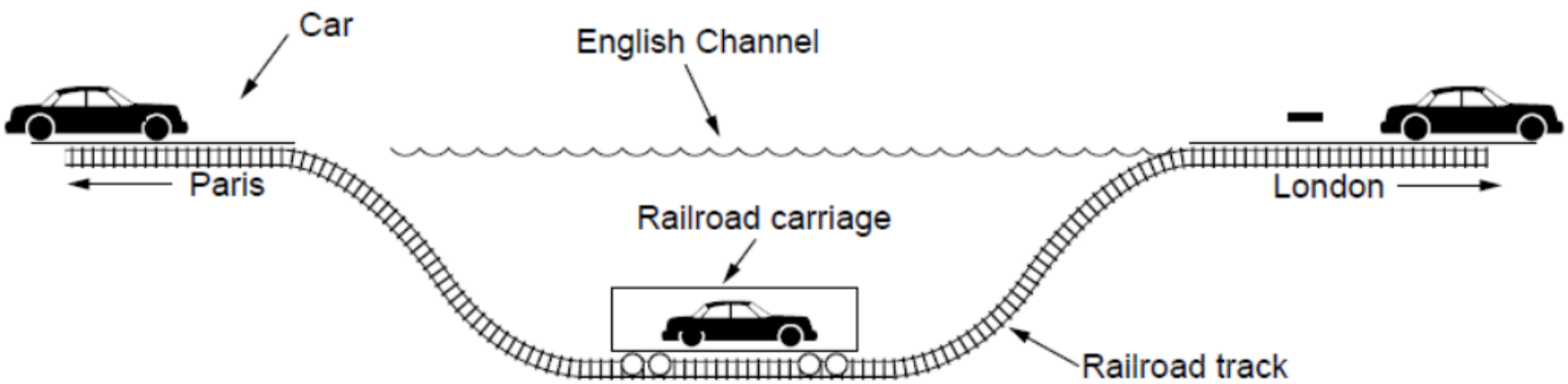
- A packet is carried whole as the payload of another protocol, so it can cross a network that could not carry it as it is.
- The middle network forwards on the outer header only. The exit router **decapsulates** and forwards the original packet unchanged. To the inner packet, the whole middle network is a single link.



Tunneling

Tunneling analogy:

- tunnel is a link; packet can only enter/exit at ends



Packet Fragmentation

- Every network limits the size of packet it can carry: its **maximum transmission unit (MTU)**, e.g. 1500 bytes of payload on Ethernet.
- Limits come from hardware, protocol standards, and the need to stop one packet from holding a link for too long.
- When a packet reaches a router whose next link has a smaller MTU, the router splits it into **fragments**, which must be **reassembled** somewhere.

Why MTUs Differ

A link's **MTU** is the largest network-layer packet (header included) it can carry in a single frame.

Where the limits come from

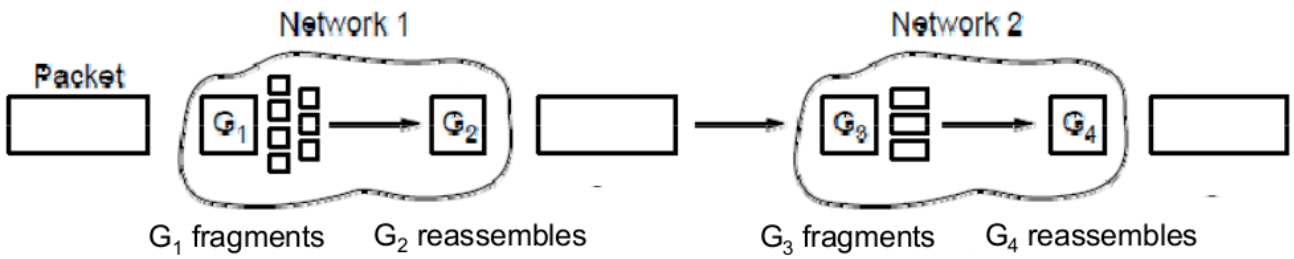
- **Link standards and hardware:** Ethernet has an MTU of 1500 bytes; 802.11 frames allow up to 2304 bytes;

Why designers keep limits small

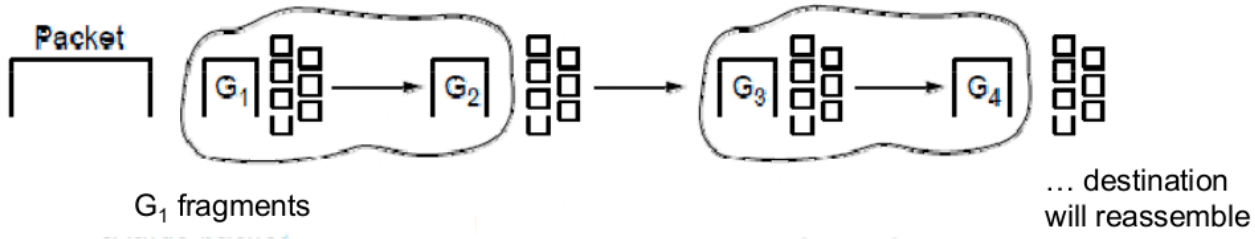
- **Fairness:** on a 64 kbps link, a 1500-byte packet occupies the line for 187.5 ms, delaying everything queued behind it.
- **Errors:** at a bit error rate of 10^{-6}, a 1500-byte packet arrives intact about 99% of the time, a 65,535-byte packet only about 59%. Any error means resending the whole packet.

Packet Fragmentation

Two Strategies:



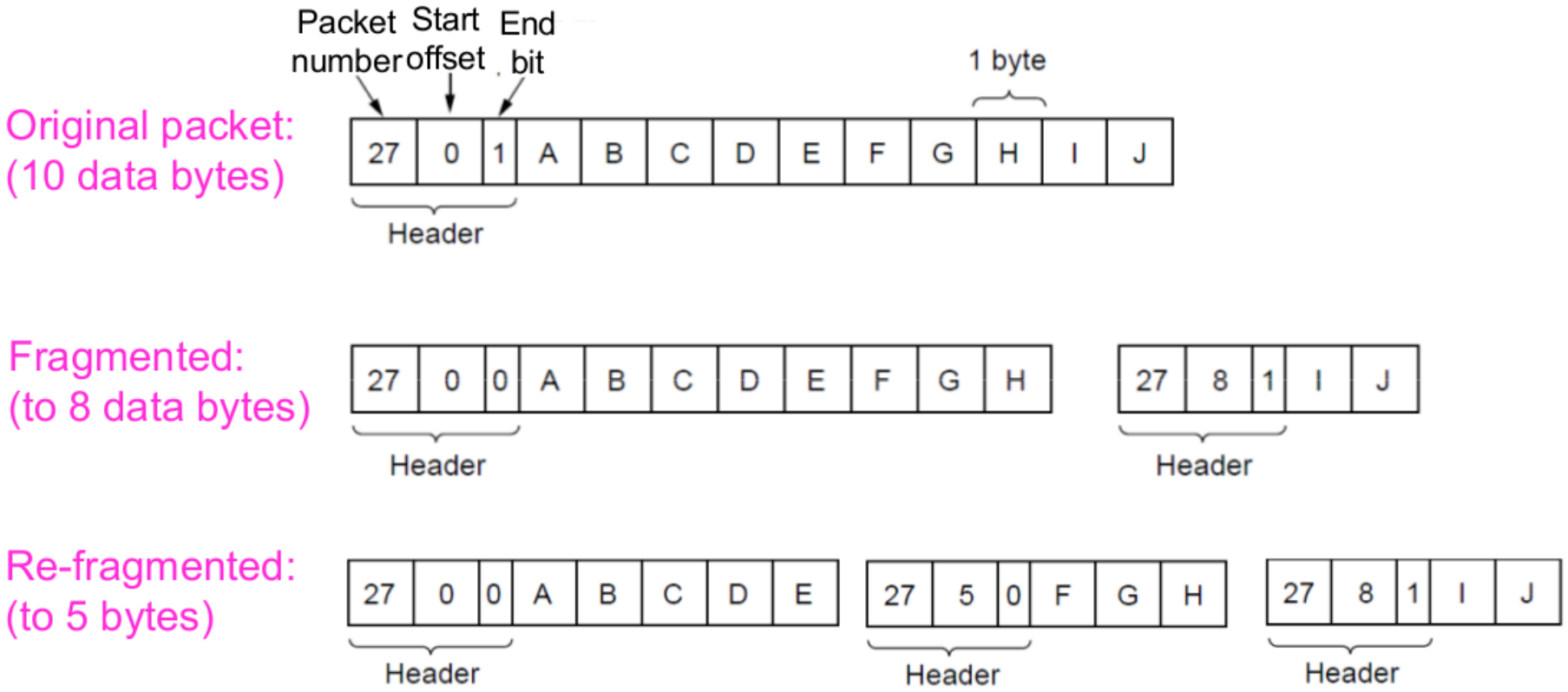
Transparent – packets fragmented / reassembled in each network



Non-transparent – fragments are reassembled at destination

Packet Fragmentation

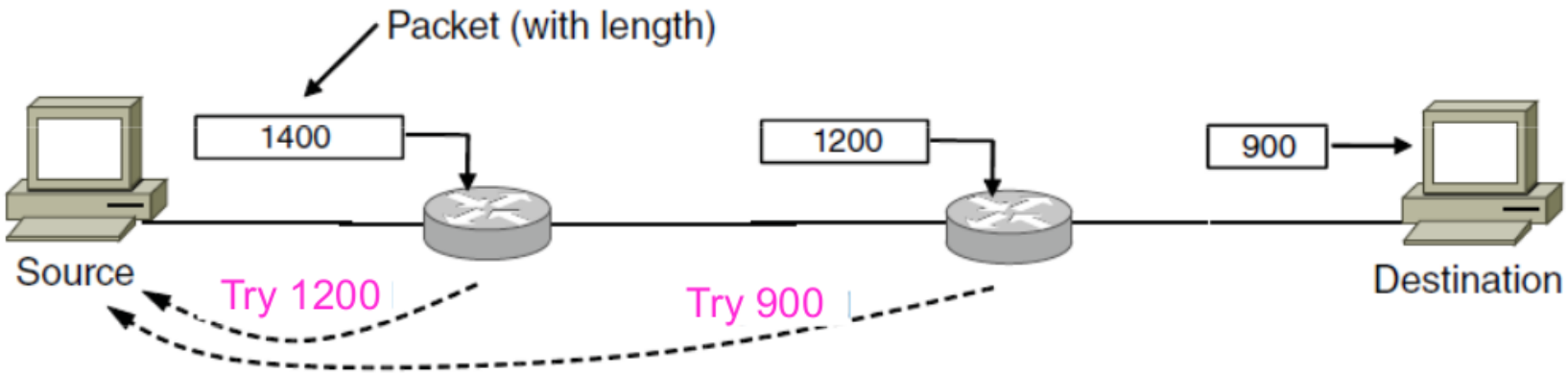
Example of IP-style fragmentation:



Path MTU Discovery

Path MTU Discovery avoids network fragmentation

- Routers return MTU (Max. Transmission Unit) to source and discard large packets



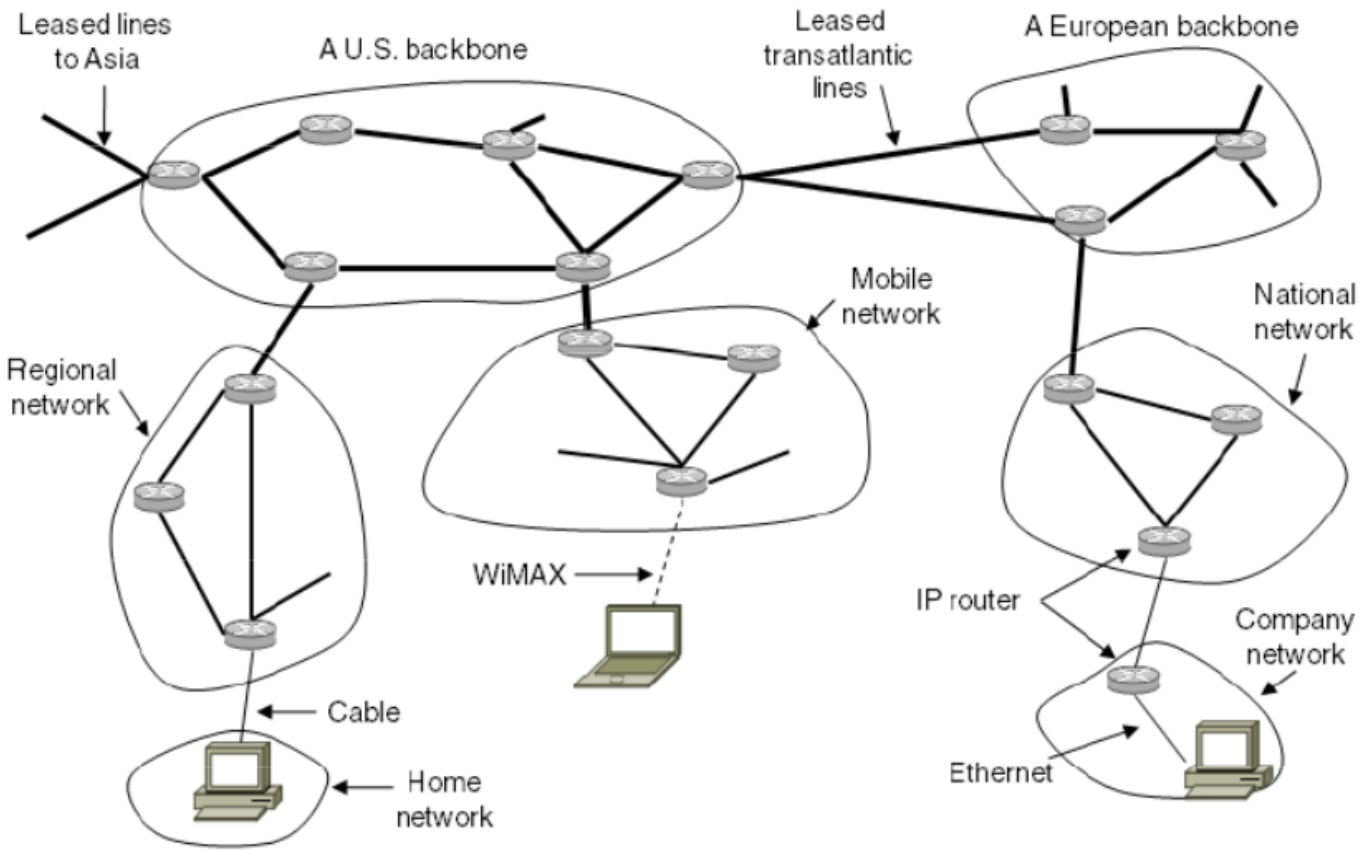
Network Layer in the Internet

IP has been shaped by guiding principles:

- Make sure it works
- Keep it simple
- Make clear choices
- Exploit modularity
- Expect heterogeneity
- Avoid static options and parameters
- Look for good design (not perfect)
- Strict sending, tolerant receiving
- Think about scalability
- Consider performance and cost

Network Layer in the Internet

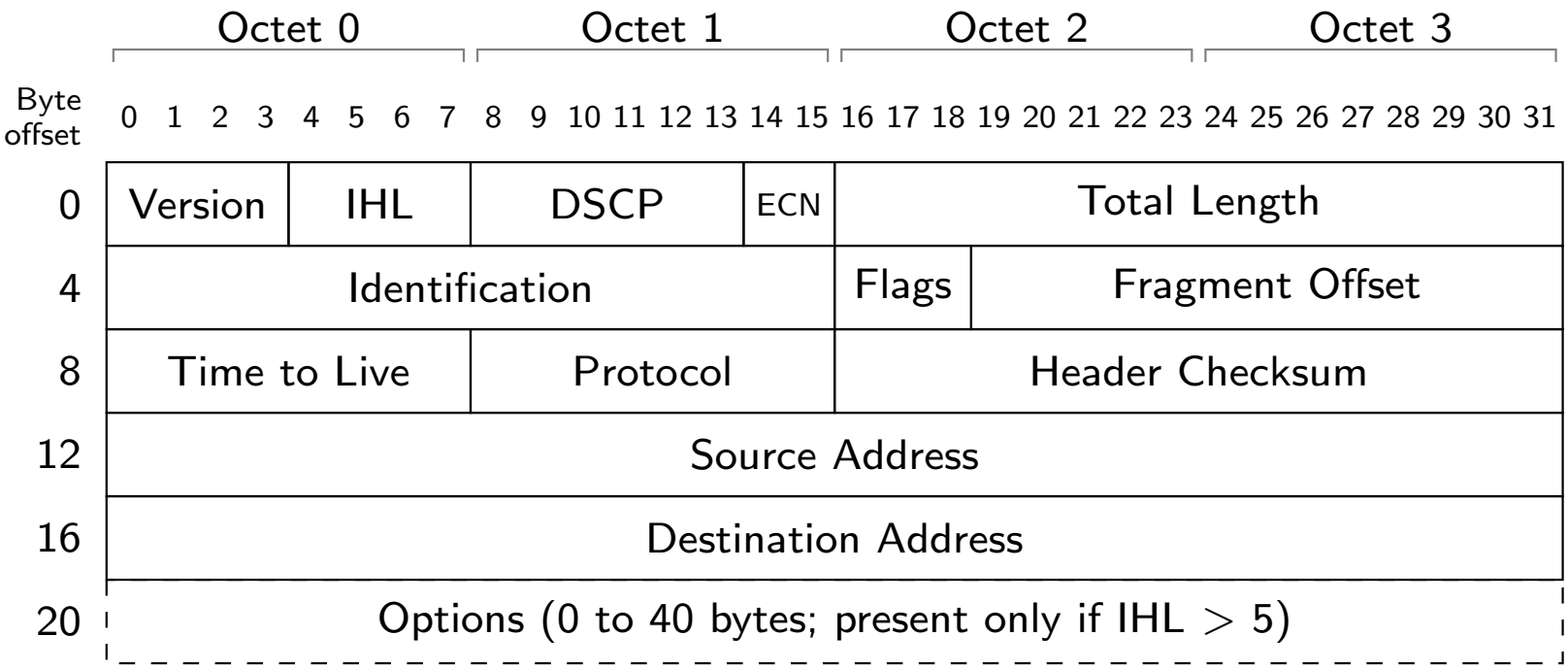
Internet is an interconnected collection of many networks that is held together by the IP protocol



Network Layer in the Internet

- The glue that holds the whole Internet together is the network layer protocol, IP (Internet Protocol).
- Unlike most older network layer protocols, IP was designed from the beginning with internetworking in mind.
- Network layer job is to provide a best-effort (i.e., not guaranteed) way to transport packets from source to destination, without regard to whether these machines are on the same network or whether there are other networks in between them.

IPv4 Protocol



- It has a 20-byte fixed part and a variable-length optional part.
- The bits are transmitted from left to right and top to bottom, with the high-order bit of the Version field going first.

IPv4 Header

- The **Version** field keeps track of which version of the protocol the datagram belongs to.
- **IHL** represents Header length in 32-bit words.
- In **Differentiated Services** top 6 bits are used to mark the packet with its service class;(ex: expedited and assured services). The bottom 2 bits are used to carry explicit congestion notification information, such as whether the packet has experienced congestion.
- The **Total length** includes everything in the datagram- both header and data. The maximum length is 65,535 bytes.
- The **Identification field** is needed to allow the destination host to determine which packet a newly arrived fragment belongs to. All the fragments of a packet contain the same Identification value.

IPv4 Header

- **DF** stands for Don't Fragment. It is an order to the routers not to fragment the packet.
- **MF** stands for More Fragments. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.
- The **Fragment offset** tells where in the current packet this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit.
- The **TtL** (Time to live) field is a counter used to limit packet lifetimes. It just counts hops. When it hits zero, the packet is discarded and a warning packet is sent back to the source host.
- **Protocol field** tells which transport process to give the packet to.
- As the header carries vital information such as addresses, it rates its own checksum for protection, using **Header checksum** field.

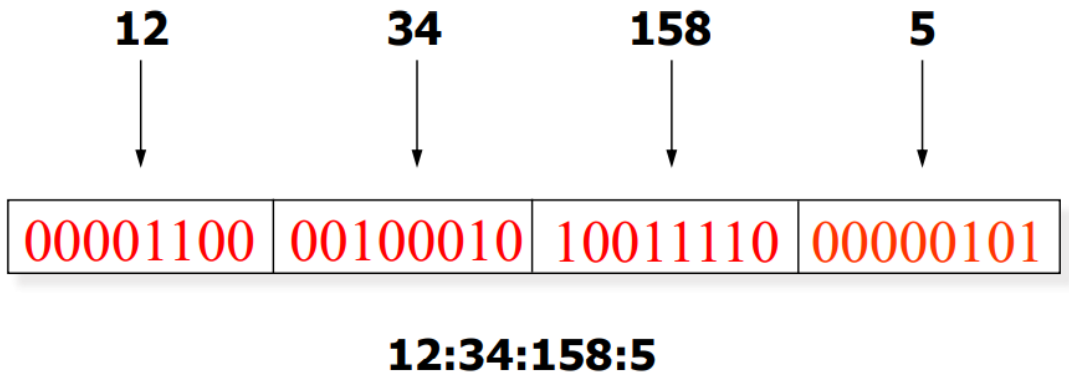
IPv4 Header Options

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Figure 5-47. Some of the IP options.

IP Addresses (IPv4)

- Every host and router on the Internet has an IP address: **a unique 32-bit number,** that can be used in the **SA** and **DA** fields of IP packets.
- IP address refers to a network interface, so if a host is on two networks, it must have two IP addresses (**routers have multiple interfaces**).



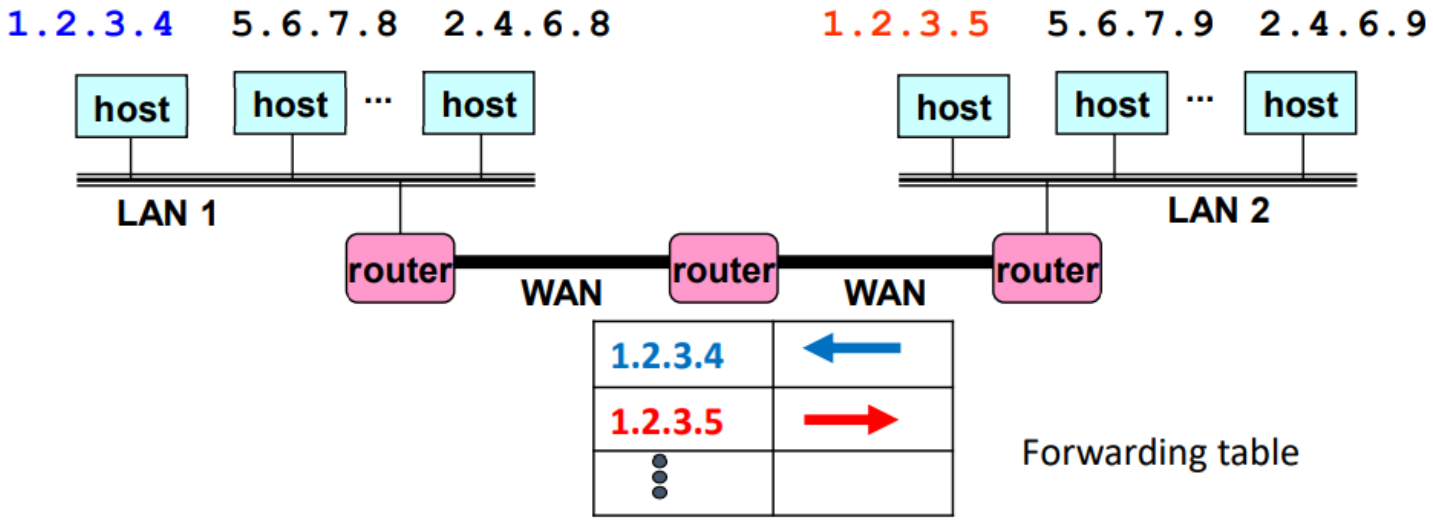
- Represented in **dotted decimal** notation.

IP Addresses (IPv4)

The Internet is an "inter-network"

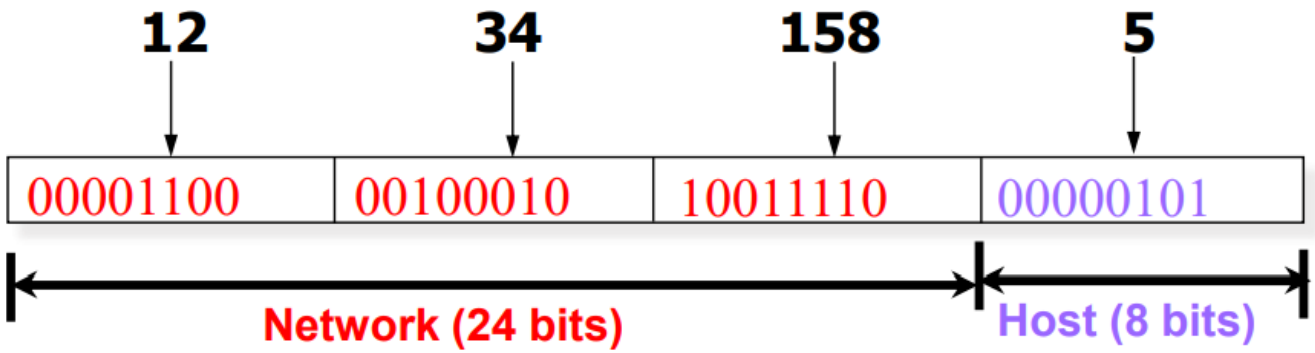
- Used to connect networks together, not hosts
- Need to address a network (i.e., group of hosts)

Suppose every host has an arbitrary address – what could be the problem?



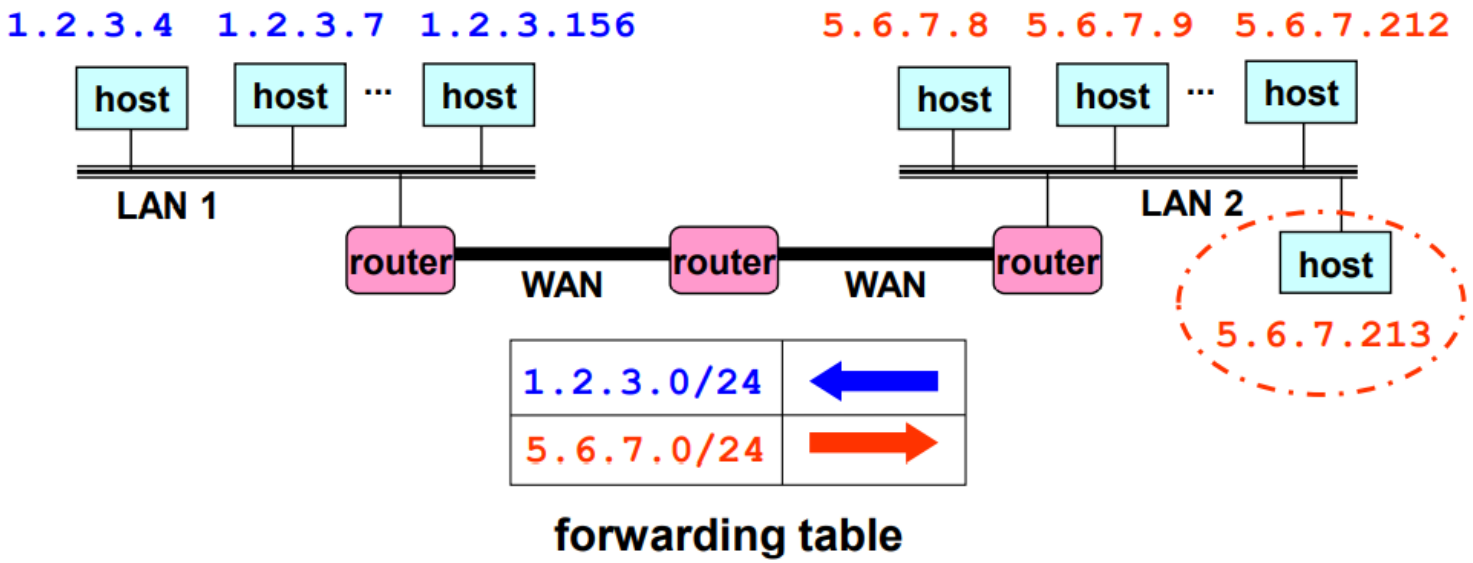
IP Addresses:Prefix

- IP addresses are hierarchical, unlike Ethernet addresses.
- Each 32-bit address is comprised of a **variable-length network portion in the top bits** and **a host portion in the bottom bits.**
- The network portion has the same value for all hosts on a single network, such as an Ethernet LAN.
- This means that a **network** corresponds to a **contiguous block of IP address space.** This block is called a **prefix.**



IP Addresses:Prefix

- Easy to add new hosts.
- No need to update the routers
  - E.g., adding a new host 5.6.7.213 on the right doesn't require adding a new forwarding-table entry



IP Addresses:Prefix

- Prefixes are written by giving the **lowest IP address in the block** and the **size of the block**.
- The size is determined by the number of bits in the network portion; the remaining bits in the host portion can vary.
- It is written after the prefix IP address as a **slash(/)** followed by the length in bits of the network portion.
- Eg: If the prefix contains 2^8 addresses, it leaves 24 bits for the **network portion**, it is written as 128.208.2.0/24.
- Since the prefix length cannot be inferred from the IP address alone, routing protocols **must carry the prefixes** to routers.
- The length of the prefix corresponds to a binary mask of **1s** in the network portion. When written out this way, it is called a **(sub)net mask**.
- It can be ANDed with the IP address to **extract only the network portion.** For our example, the (sub)net mask is 255.255.255.0.

IP Addresses

- **Address** - The unique number ID assigned to one host or interface in a network.
- A **network** corresponds to a **contiguous block of IP address space.** This block is called a **prefix.**
- **Subnet** - A portion of a network that shares a particular subnet address.
- **Subnet mask** - A 32-bit combination used to describe which portion of an address refers to the subnet and which part refers to the host.
- **Interface** - A network connection.

Obtaining IP Addresses

- Network numbers are managed by a non-profit corporation called **ICANN (Internet Corporation for Assigned Names and Numbers)**, to avoid conflicts.
- In turn, **ICANN** has delegated parts of the address space to various **regional authorities**, which dole out IP addresses to ISPs and other companies.
- Internet Service Providers (ISPs) allocate address blocks to their customers, who may, in turn, allocate to their customers..

Subnetting: the Problem

- Routing by prefix treats every address with the same prefix as being on one network.
- A university receives a /16 block and uses it for the Computer Science Dept.'s Ethernet. Routers outside send every address in the /16 to the university, and the university router delivers all of them onto the CS Ethernet.
- A year later, the Electrical Engineering Dept. wants to get on the Internet, and the Art Dept. soon follows, each with its own Ethernet. **What addresses should they use?**

Subnetting: the Problem

- Routing by prefix treats every address with the same prefix as being on one network.
- A university receives a /16 block and uses it for the Computer Science Dept.'s Ethernet. Routers outside send every address in the /16 to the university, and the university router delivers all of them onto the CS Ethernet.
- A year later, the Electrical Engineering Dept. wants to get on the Internet, and the Art Dept. soon follows, each with its own Ethernet. **What addresses should they use?**
- Addresses taken from the /16 would be delivered onto the CS Ethernet, not theirs. Getting a new block for each department is the other option (next slide).

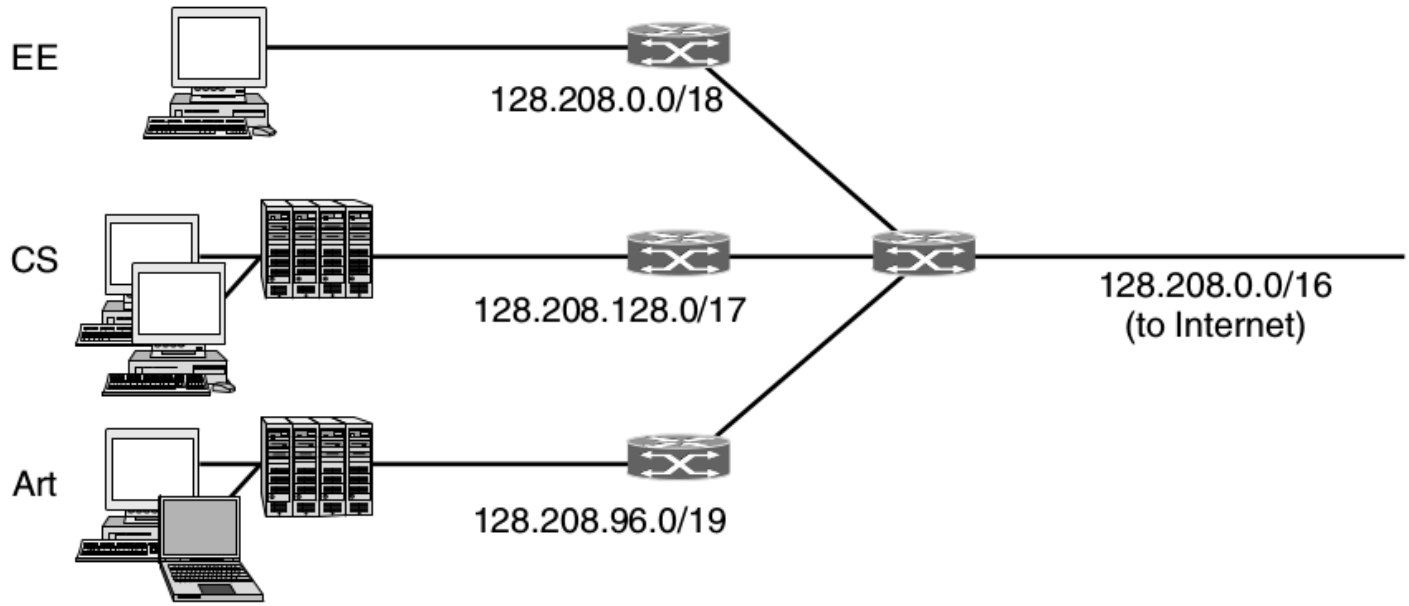
subnets

- Getting further blocks requires going outside the university: to its ISP or a regional registry and may be expensive or inconvenient.
- Moreover, the /16 already allocated has enough addresses for over **60,000** hosts.
- It might be intended to allow for significant growth, but until that happens, it is wasteful to allocate further blocks of IP addresses to the same university. **A different organization is required.**

subnets

- The solution is to allow the block of addresses to be split into several parts for internal use as multiple networks, while still acting like a single network to the outside world.
- This is called **subnetting** and the networks (such as Ethernet LANs) that result from dividing up a larger network are called subnets.

Subnetting: Splitting the University Block



- The single **/16** has been split into pieces of different sizes: half of the block (**a /17**) is allocated to the Computer Science Dept., a quarter (**a /18**) to the Electrical Engineering Dept., and an eighth (**a /19**) to the Art Dept. The remaining eighth is unallocated.

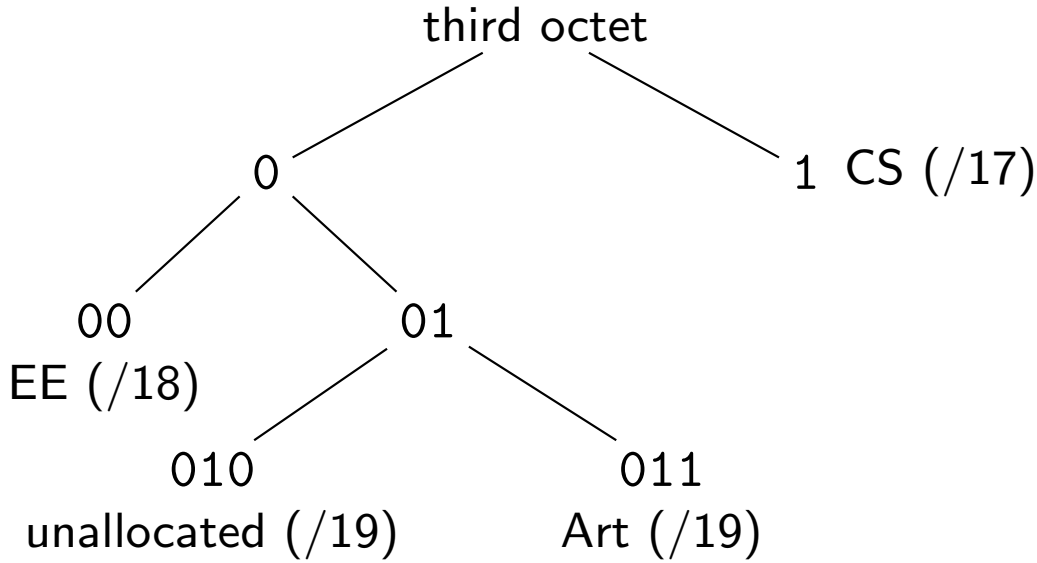
Subnetting: How Many Subnet Bits?

Each department's share of 128.208.0.0/16 fixes how many subnet bits it needs:

Dept.	Share	Subnet bits	Prefix	Host bits	Addresses
CS	1/2	1	/17	15	32,768
EE	1/4	2	/18	14	16,384
Art	1/8	3	/19	13	8,192

- A share of $1/2^k$ needs k subnet bits: the size fixes the *length* of each bit pattern.
- The *values* are chosen by the network administrator, with one rule: no pattern may be the start of another. (If CS had 1 and EE had 10, every EE address would also lie in CS's block.)
- Allocate the largest block first, then fit each smaller block into a free part of the space.

Subnetting: Choosing the Bit Patterns



Dept.	**Prefix**	**Share**	**Third octet**
Electrical Eng.	128.208.0.0/18	1/4	0 to 63
(unallocated)	128.208.64.0/19	1/8	64 to 95
Art	128.208.96.0/19	1/8	96 to 127
Computer Science	128.208.128.0/17	1/2	128 to 255

Other layouts are equally valid, e.g. CS = 0, EE = 10, Art = 110.

Subnetting: Prefixes in Binary

The university's block 128.208.0.0/16 after the split, in binary (the bar | marks the end of each department's prefix):

Dept.	**Address bits**	**Prefix**
Computer Science	10000000 11010000 1|xxxxxxx xxxxxxxx	/17
Electrical Eng.	10000000 11010000 00|xxxxxx xxxxxxxx	/18
Art	10000000 11010000 011|xxxxx xxxxxxxx	/19

- Each address now has three levels: the **network** part (the first 16 bits, 128.208, assigned from outside), the **subnet** bits (chosen inside the university), and the **host** bits (the x's).
- The subnet part has a different length in each department, so each has its own mask: /17 is 255.255.128.0, /18 is 255.255.192.0, and /19 is 255.255.224.0.
- Routers outside the university see only 128.208.0.0/16.

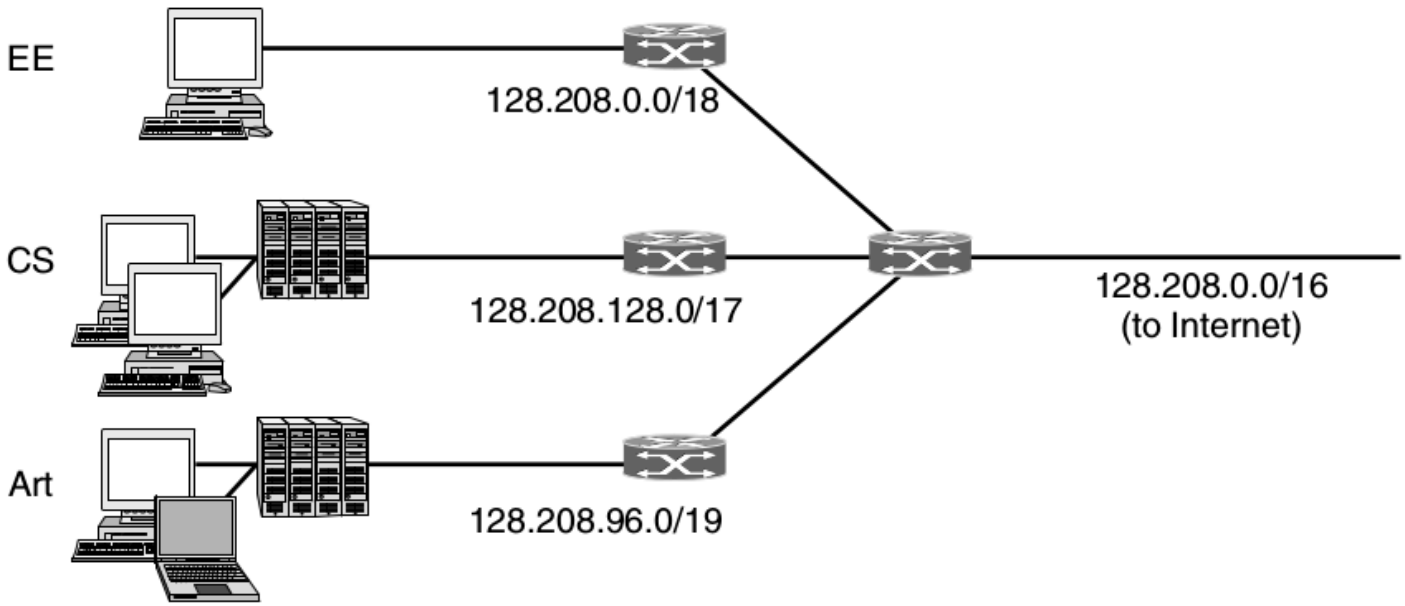
Subnetting: One Mask per Subnet

Dept.	Prefix	Mask	Third octet of mask
CS	/17	255.255.128.0	10000000 = 128
EE	/18	255.255.192.0	11000000 = 192
Art	/19	255.255.224.0	11100000 = 224
(unallocated)	/19	255.255.224.0	11100000 = 224

- The mask has one 1 in the third octet for each subnet bit; 255.255 covers the university's /16.
- Every host in a department uses that department's mask, set manually or by DHCP.
- The same mask does not mean the same subnet: Art and the unallocated block are both /19 but have different prefixes (128.208.96.0 and 128.208.64.0).
- Using different mask lengths within one block is called **variable-length subnet masking (VLSM)**. The router stores each subnet's mask in its table entry.

subnets

- When a packet comes into the main router, how does the router know which subnet to give it to?
- One way would be for each router to have a table with 65,536 entries telling it which outgoing line to use for each host on campus.
- But this would undermine the main scaling benefit we get from using a hierarchy. Instead, the routers simply need to know the subnet masks for the networks on campus.



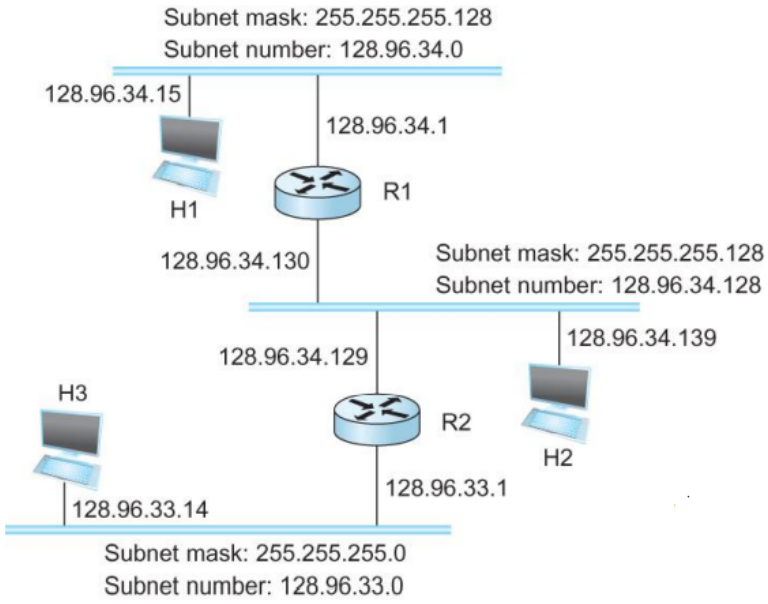
subnets

- When a packet arrives, the router looks at the destination address of the packet and checks which subnet it belongs to.
- The router can do this by **ANDing** the **destination address** with the **mask for each subnet** and checking to see if the result is the corresponding prefix.
- For example, consider a packet destined for IP address 128.208.2.151. To see if it is for the Computer Science Dept., we AND with 255.255.128.0 to take the first 17 bits (which is 128.208.0.0) and see if they match the prefix address (which is 128.208.128.0). They do not match.

subnets

- Checking the first 18 bits for the Electrical Engineering Dept., we get 128.208.0.0 when ANDing with the subnet mask. This does match the prefix address, so the packet is forwarded onto the interface which leads to the Electrical Engineering network.
- The subnet divisions can be changed later if necessary, by updating all subnet masks at routers inside the university.

Sub-netting



Forwarding table at R1

SubnetNumber	SubnetMask	NextHop
128.96.34.0	255.255.255.128	Interface 0
128.96.34.128	255.255.255.128	Interface 1
128.96.33.0	255.255.255.0	R2

```
D = destination IP address
for each entry <SubnetNum, SubnetMask, NextHop>:
  D1 = SubnetMask AND D
  if D1 = SubnetNum:
    if NextHop is an interface:  deliver directly to destination
    else:  deliver to NextHop (a router)
if no entry matches:  deliver to the default router
```

Understanding Subnetting

An organisation is given the block 172.16.0.0/16. Splitting it with a longer prefix is **subnetting**.

Question: with a /21 mask (255.255.248.0), how many subnets are there, and how many hosts can each hold?

Understanding Subnetting

An organisation is given the block 172.16.0.0/16. Splitting it with a longer prefix is **subnetting**.

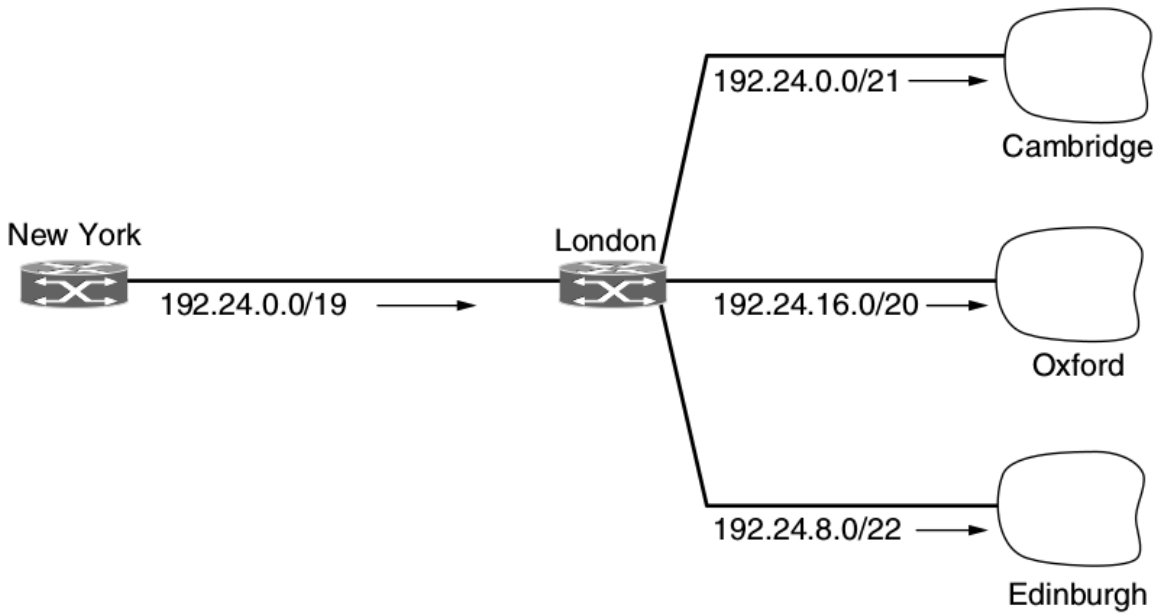
Question: with a /21 mask (255.255.248.0), how many subnets are there, and how many hosts can each hold?

- Subnet bits: $21 - 16 = 5$, so $2^5 = 32$ subnets.
- Host bits: $32 - 21 = 11$, so $2^{11} = 2048$ addresses per subnet, of which $2048 - 2 = 2046$ can be hosts (all-zeros and all-ones host parts are reserved).
- Third octet of the mask: $248 = 11111000_2$, so subnets start every $256 - 248 = 8$: 172.16.0.0/21, 172.16.8.0/21, . . . , 172.16.248.0/21.
- First subnet: hosts 172.16.0.1 to 172.16.7.254; broadcast 172.16.7.255.

CIDR: Classless InterDomain Routing

Sub-netting allows addresses to be used efficiently, but there is still a problem that remains: routing table explosion.

Example:



CIDR: Classless InterDomain Routing

- Routers in organizations at the edge of a network, such as a university, need to have an entry for each of their sub-nets, telling the router which line to use to get to that network.
- For routes to destinations outside of the organization, they can use the simple default rule of sending the packets on the line toward the ISP that connects the organization to the rest of the Internet.
- Routers in ISPs and backbones in the middle of the Internet must know which way to go to get to every network and no simple default will work.
- These core routers are said to be in the **default-free zone** of the Internet.
- **Challenges:** Large tables, large lookup time, routing table population.

CIDR: Classless InterDomain Routing

- **Solution:** Combine multiple small prefixes into a single larger prefix. This process is called route aggregation.
- This design works with sub-netting and is called CIDR.

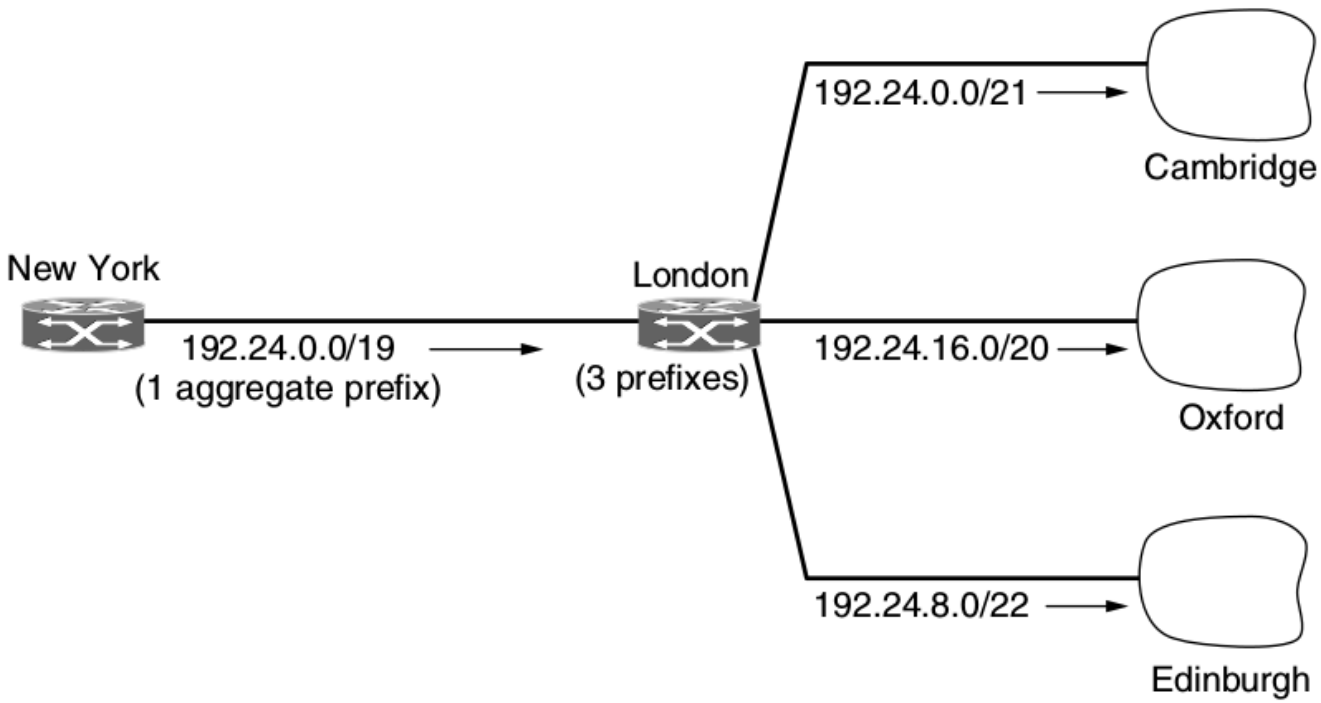
CIDR: Example

- Let us consider an example in which a block of 8192 IP addresses is available starting at 194.24.0.0.
- Suppose, Cambridge University needs 2048 addresses. Next, Oxford University asks for 4096 addresses. Finally, the University of Edinburgh asks for 1024 addresses.

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

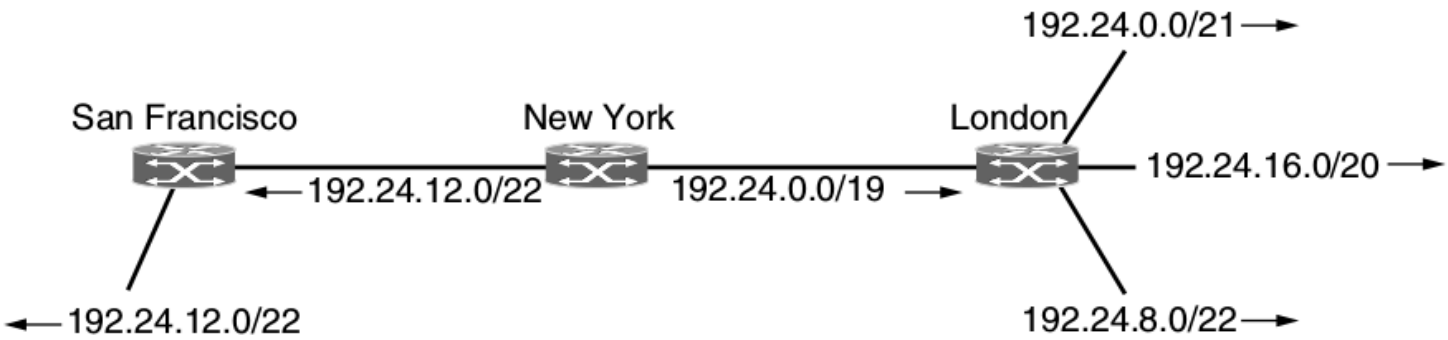
CIDR

- All of the routers in the default-free zone are now told about the IP addresses in the three networks.
- All of the IP addresses in the three prefixes should be sent from New York (or the U.S. in general) to London.



CIDR

- Prefixes are allowed to overlap. The rule is that packets are sent in the direction of the most specific route, or the longest matching prefix.



Calculate for IP:192.24.14.5?

Historical Note: Classful Addressing (before 1993)

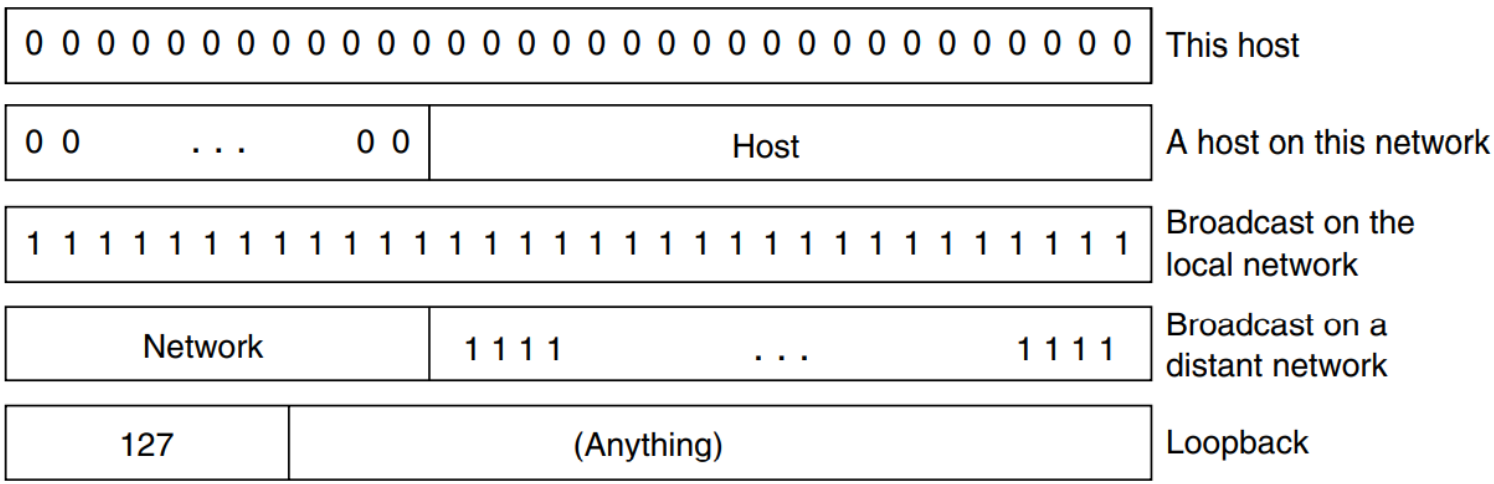
Originally the prefix length was fixed by the first bits of the address:

Class	Leading bits	Prefix	Range
A	0	/8	0.0.0.0 to 127.255.255.255
B	10	/16	128.0.0.0 to 191.255.255.255
C	110	/24	192.0.0.0 to 223.255.255.255
D	1110	multicast	224.0.0.0 to 239.255.255.255
E	1111	reserved	240.0.0.0 to 255.255.255.255

- **Wasted addresses:** only three block sizes. A network of 255 hosts did not fit a class C (at most 254 hosts), so it needed a class B: 65,534 host addresses for 255 hosts.
- **Large routing tables:** every network was a separate entry, with no way to aggregate.
- **CIDR (1993)** replaced classes with prefixes of any length. Classes D and E survive as the multicast (224.0.0.0/4) and reserved (240.0.0.0/4) blocks, and "class C" is still used loosely to mean a /24.

Special Addresses

- The IP address 0.0.0.0, the lowest address, is used by hosts when they are being booted. It means "this network" or "this host."



Network Address Translation

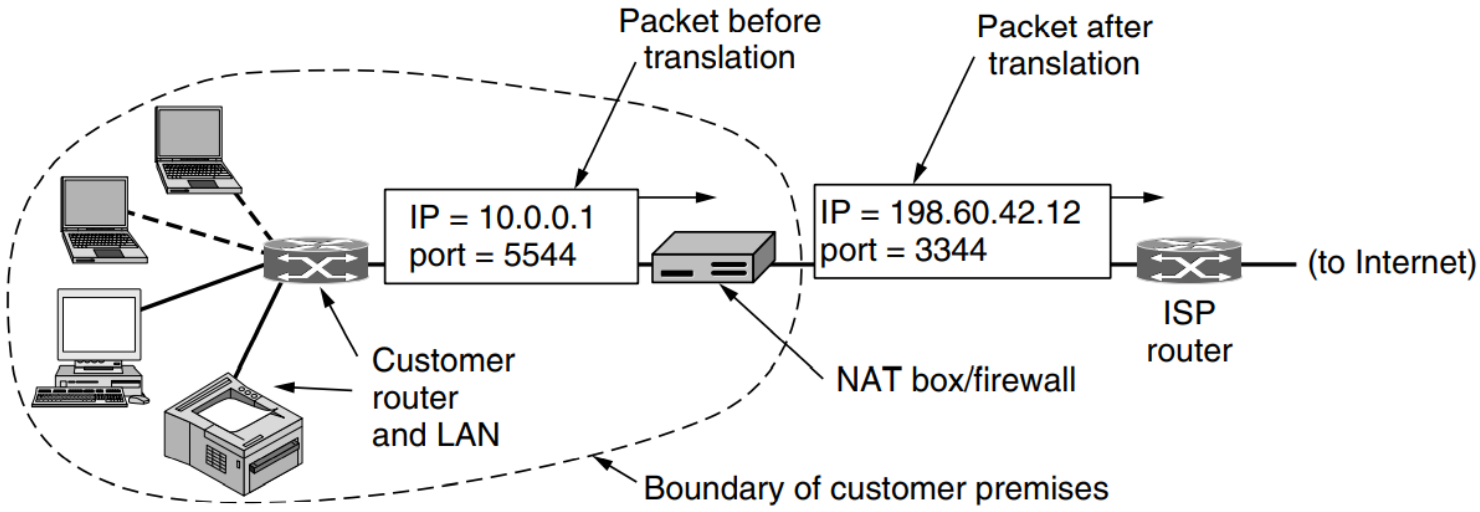
- IP addresses are scarce.
- An ISP might have a /16 address, giving it 65,534 usable host numbers. If it has more customers than that, it has a problem.
- This scarcity has led to techniques to use IP addresses sparingly.
- One approach is to dynamically assign an IP address to a computer when it is on and using the network.
- With the techniques we have seen so far, **each computer must have its own IP address all day long.**

Network Address Translation

- The basic idea behind NAT is for the ISP to assign each home or business a single IP address (or at most, a small number of them) for Internet traffic.
- Within the customer network, every computer gets a unique IP address, which is used for routing intramural traffic.
- However, just before a packet exits the customer network and goes to the ISP, an address translation from the unique internal IP address to the shared public IP address takes place.
- This translation makes use of three ranges of IP addresses that have been declared as private.
- **No packets containing these addresses may appear on the Internet**. The three reserved ranges are:

10.0.0.0 – 10.255.255.255/8 (16,777,216 hosts)
172.16.0.0 – 172.31.255.255/12 (1,048,576 hosts)
192.168.0.0 – 192.168.255.255/16 (65,536 hosts)

NAT operation



- NAT designers observed that most IP packets carry either TCP or UDP payloads.
- Whenever an outgoing packet enters the NAT box, the 10.x.y.z source address is replaced by the customer's true IP address.
- The TCP Source port field is replaced by an index into the NAT box's 65,536-entry translation table.
- This table entry contains the original IP address and the original source port.

Internet Control Protocols

- IP is used for data transfer. Internet has several companion control protocols that are used in the network layer.
- They include ICMP, ARP, and DHCP.

Internet Control Message Protocol

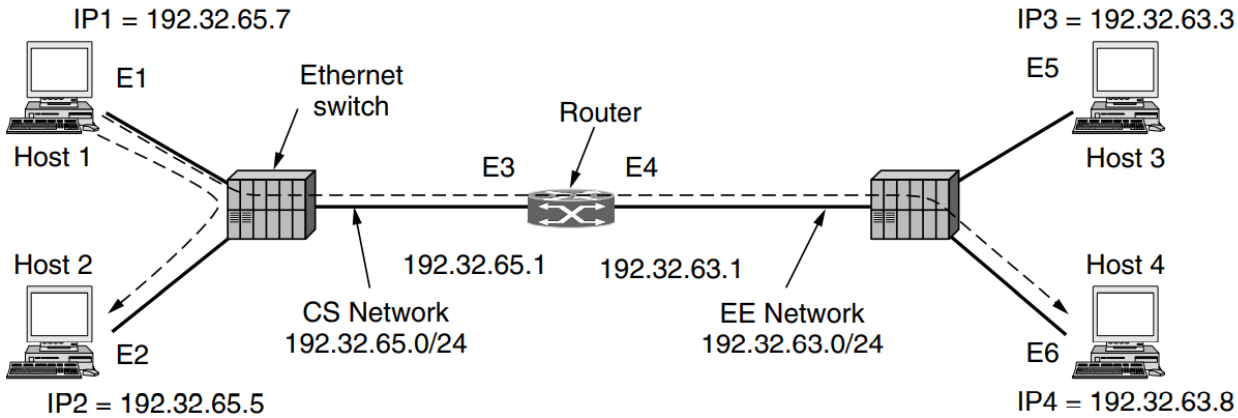
- When something unexpected occurs during packet processing at a router, the event is reported to the sender by ICMP.
- ICMP is also used to test the Internet.
- Each ICMP message type is carried encapsulated in an IP packet.

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

All messages can be found at:
www.iana.org/assignments/icmp-parameters.

Address Resolution Protocol

- IP addresses are not sufficient for sending packets in a LAN(Ethernet/WiFi).
- Data link layer NICs such as Ethernet cards do not understand Internet addresses.
- How do IP addresses get mapped onto data link layer addresses, such as Ethernet?



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

Address Resolution Protocol

Let us assume the sender(H1) knows the name of the intended receiver(H2):

- Step 1: To find the IP address for host 2. This lookup is done by DNS.
- Step 2: Build a packet with 192.32.65.5 in the Destination address field and gives it to the IP software to transmit.
- Step 3:Host 1 outputs a broadcast packet onto the Ethernet asking who owns IP address 192.32.65.5(H2).
- Step 4: Host 2 alone will respond with its Ethernet address (E2)

Optimizations: Caching, gratuitous ARP.

Address Resolution Protocol

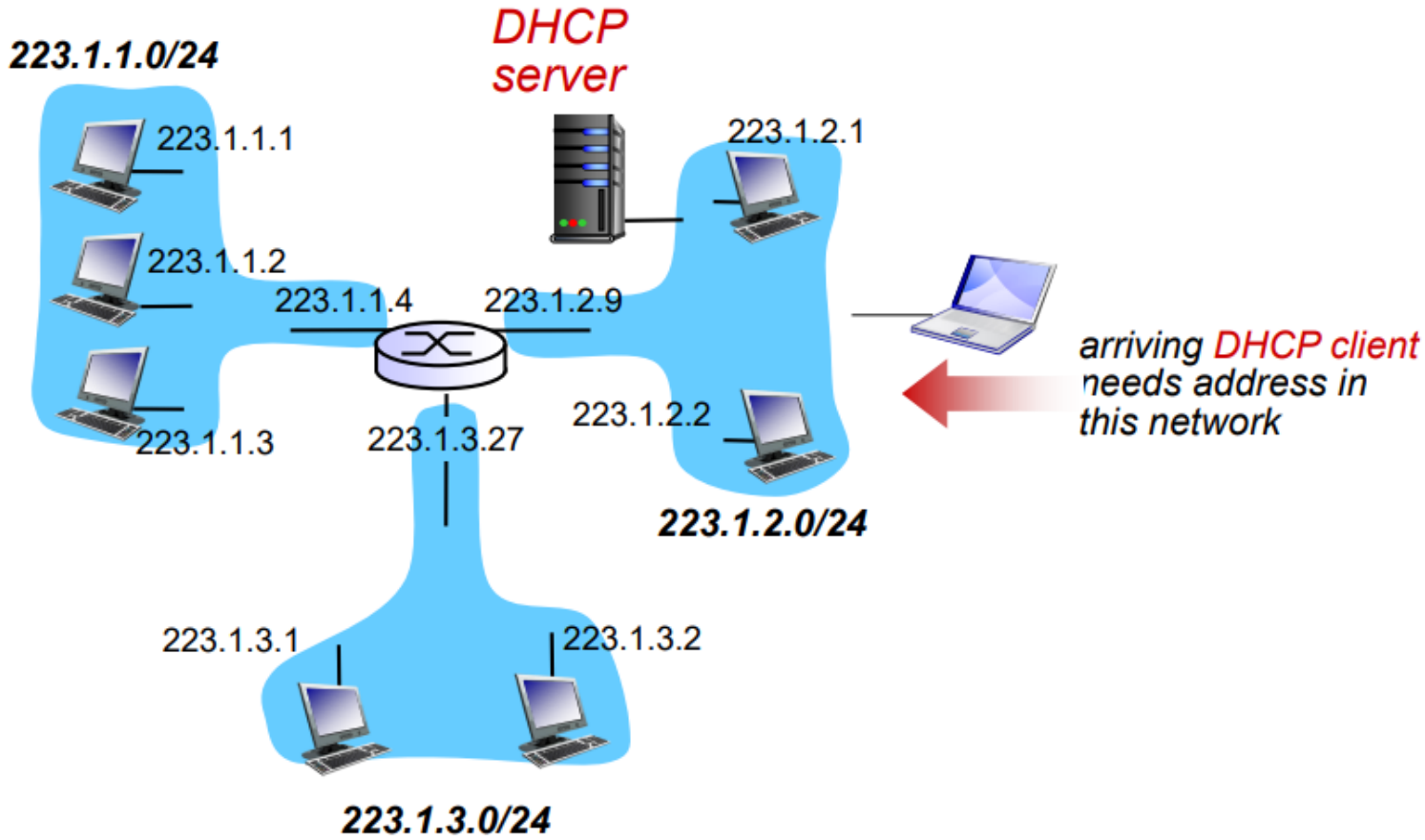
- How to deal when a host in on another network?
- Use the default gateway or proxy ARP.

Dynamic Host Configuration Protocol

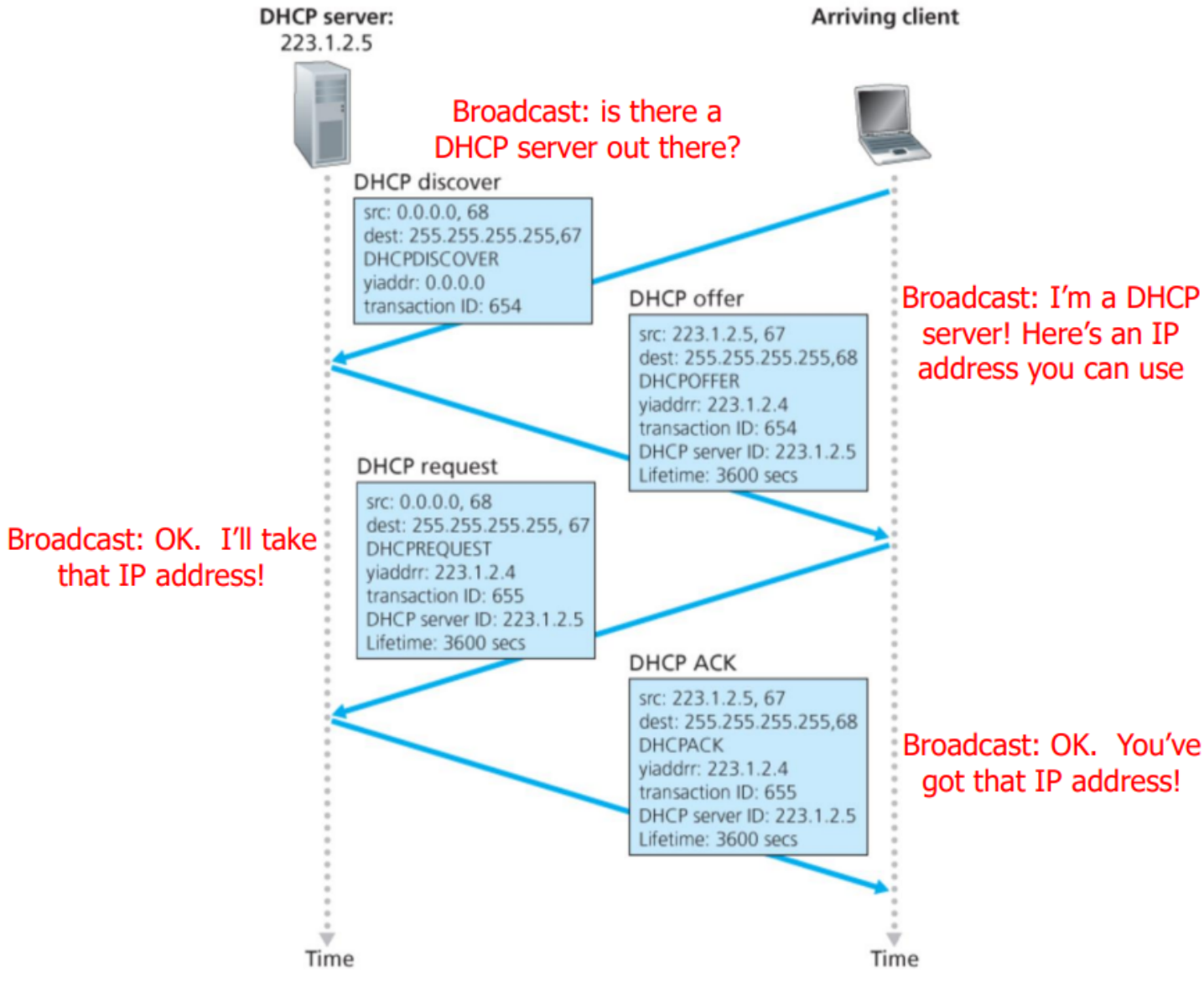
- ARP (as well as other Internet protocols) makes the assumption that hosts are configured with some basic information, such as their own IP addresses. How do hosts get this information?

DHCP overview:
host broadcasts "DHCP discover" msg [optional]
DHCP server responds with "DHCP offer" msg [optional]
host requests IP address: "DHCP request" msg
DHCP server sends address: "DHCP ack" msg

DHCP Client-Server Scenario



DHCP Operation



DHCP Operation

- DHCP server is responsible for providing configuration information to hosts
- There is at least one DHCP server for an administrative domain
- DHCP server maintains a pool of available addresses
- Newly booted or attached host sends DHCPDISCOVER message to a special IP address (255.255.255.255)
- DHCP relay agent unicasts the message to DHCP server and waits for the response

Thank you